

Ficha 2

$$\textcircled{9} \text{ a) } \int (n+1) \operatorname{sen} n \, dn = \int v \cdot v' = v \cdot v - \int v \cdot v' = -\cos n \cdot (n+1) + \int \cos n$$

$$v' = \operatorname{sen} n \quad v = n+1 \quad = -\cos n(n+1) + \operatorname{sen} n + C, C \in \mathbb{R}$$

$$v = -\cos n \quad v' = 1$$

$$\text{b) } \int n \cos(n) \, dn = \int v \cdot v' = v \cdot v - \int v' \cdot v = \operatorname{sen}(n) \cdot n - \int 1 \cdot \operatorname{sen} n \, dn$$

$$v' = \cos n \quad v = n \quad = n \operatorname{sen} n + \cos n$$

$$v = \operatorname{sen} n \quad v' = 1$$

$$\text{c) } \int n^2 \cos(n) \, dn = \int v \cdot v' = v \cdot v - \int v' \cdot v = n^2 \operatorname{sen} n - \int 2n \operatorname{sen} n$$

$$v' = \cos n \quad v = n^2 \quad = n^2 \operatorname{sen} n - 2 \int n \operatorname{sen} n = \dots$$

$$v = \operatorname{sen} n \quad v' = 2n$$

$$10) a) \int x \sqrt{x+1}$$

$$\int (t^2-1) \cdot t \cdot 2t \, dt = \int t^3 - t \cdot 2t \, dt = \int 2t^4 - 2t^2 \, dt$$

$$t = \sqrt{x+1} \Leftrightarrow t^2 = x+1 \Leftrightarrow x = t^2-1$$

$$dx = 2t \, dt$$

$$= 2 \left(\frac{t^5}{5} - \frac{t^3}{3} \right)$$

$$= 2 \left(\frac{\sqrt{x+1}^5}{5} - \frac{\sqrt{x+1}^3}{3} \right) + C, C \in \mathbb{R}$$

$$b) \int \frac{x}{1+\sqrt{x}} \, dx \longrightarrow \int \frac{(t-1)^2}{t} \cdot 2(t-1) \, dt = 2 \int \frac{(t-1)^3}{t} \, dt$$

$$t = 1 + \sqrt{x} \Leftrightarrow (t-1)^2 = x \quad = 2 \int \frac{(t-1)(t^2-2t-1)}{t} \, dt$$

$$dx = 2(t-1) \, dt$$

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$$⑥ \int \frac{1}{x^2+1} \text{ que se anula em } x=2?$$

$$\int \frac{1}{x^2+1} \, dx = \int x^{-2} + 1 \, dx = -\frac{1}{2} \int -2 x^{-2} \, dx + \int 1 \, dx = -\frac{1}{2} \left(\frac{x^{-1}}{-1} \right) + x + C$$

$$= -\frac{1}{2} \left(-\frac{1}{n} \right) + n + C = \frac{1}{2n} + n + C, C \in \mathbb{R}$$

$$\frac{1}{2 \cdot 2} + 2 + C = 0 \quad (=) \quad \frac{1}{4} + 2 = -C \quad (=) \quad \frac{1}{4} + \frac{8}{4} = -C \quad (=) \quad \frac{9}{4} = -C \rightarrow \frac{1}{2n} + n - \frac{9}{4}$$

$$\textcircled{7} \quad \int \frac{3 \cos(\ln n)}{n} \quad = 2 \quad \text{em} \quad n = 1$$

$$\int \frac{3 \cos(\ln n)}{n} \, dn = 3 \int \cos(\ln n) \cdot \frac{1}{n} \, dn = 3 \sin(\ln n) + C$$

$$3 \sin(\ln(1)) + C = 2 \quad (=) \quad 3 \sin(0) - 2 = -C \quad (=) \quad C = 2 \rightarrow 3 \sin(\ln n) + 2$$

$$\textcircled{1} \quad \text{a) } \int 3n^2 + 5n + 7 \, dn = \int 3n^2 \, dn + \int 5n \, dn + \int 7 \, dn =$$

$$= n^3 + \frac{5n^2}{2} + 7n + C, C \in \mathbb{R}$$

$$b) \int \sqrt[3]{x} \, dx = \int x^{1/3} \, dx = \frac{x^{-\frac{2}{3}}}{-\frac{2}{3}} + C = -\frac{3}{2} \cdot \frac{1}{\sqrt[3]{x^2}} + C$$

$$c) \int (x^3 + 1)^2 \, dx = \int x^6 + 2x^3 + 1 \, dx = \frac{x^7}{7} + \frac{2x^4}{4} + x + C, \quad C \in \mathbb{R}$$

$$d) \int \frac{\operatorname{arctg} x}{1+x^2} \, dx = \int \underbrace{\frac{1}{1+x^2}}_{u'} \underbrace{\operatorname{arctg} x}_u \, dx = \left(\frac{\operatorname{arctg} x}{2} \right)^2 + C, \quad C \in \mathbb{R}$$

$$e) \int \frac{3x^2}{1+x^3} \, dx \stackrel{\frac{u'}{u}}{=} \ln |1+x^3| + C, \quad C \in \mathbb{R}$$

$$f) \int \frac{1}{x^7} \, dx = \int x^{-7} \, dx = \frac{x^{-6}}{-6} + C = -\frac{1}{6x^6} + C, \quad C \in \mathbb{R}$$

$$\begin{aligned} g) \int \frac{x+1}{2+4x^2} \, dx &= \frac{1}{2} \int \frac{x+1}{1+2x^2} \, dx = \frac{1}{2} \int \frac{x}{1+2x^2} \, dx + \frac{1}{2} \int \frac{1}{1+2x^2} \, dx = \\ &= \frac{1}{2} \cdot \frac{1}{4} \int \frac{4x}{1+2x^2} \, dx + \frac{1}{2} \cdot \frac{1}{2} \int \frac{2}{1+2x^2} \, dx = \frac{1}{8} \ln(1+2x^2) + \frac{1}{4} \operatorname{arctg}(2x) \end{aligned}$$

$$h) \int 4x^3 \cos x^4 \, dx = \sin(x^4) + C, \quad C \in \mathbb{R}$$

$$\begin{aligned}
 i) \int \frac{x}{\sqrt{1-x^2}} dx &= \int x \cdot \frac{1}{\sqrt{1-x^2}} dx = \int x \cdot (1-x^2)^{-1/2} = -\frac{1}{2} \int -2x \cdot (1-x^2)^{-1/2} dx \\
 &= -\frac{1}{2} \frac{(1-x^2)^{1/2}}{\frac{1}{2}} + C = -(1-x^2)^{1/2} = -\sqrt{1-x^2}
 \end{aligned}$$

$$j) \int \underbrace{\sin x}_{-u'} \underbrace{\cos^5 x}_{u^5} dx = -\frac{\cos^6 x}{6}$$

$$k) \int \operatorname{tg} x dx = \int \frac{\sin x}{\cos x} dx = -\ln |\cos x| + C, \quad C \in \mathbb{R}$$

$$l) \int \frac{\ln x}{x} dx = \int \frac{1}{x} \ln x dx = \frac{\ln^2 x}{2} + C, \quad C \in \mathbb{R}$$

$$m) \int e^{\overbrace{\operatorname{tg} x}^u} \underbrace{\sec^2 x}_{u'} dx = e^{\operatorname{tg} x}$$

$$n) \int x \cdot 7^{x^2} dx = \frac{1}{2} \int 2x \cdot 7^{x^2} = \frac{7^{x^2}}{2 \ln(7)}$$

$$o) \int \sin(\sqrt{2} x) dx = \frac{1}{\sqrt{2}} \int \sqrt{2} \sin(\sqrt{2} x) dx = \frac{-\cos(\sqrt{2} x)}{\sqrt{2}} + C, C \in \mathbb{R}$$

$$p) \int \frac{x^2 + 1}{x} dx = \int \frac{x^2}{x} + \frac{1}{x} dx = \int x + \frac{1}{x} dx = \frac{x^2}{2} + \ln|x| + C, C \in \mathbb{R}$$

$$q) \int \frac{x}{(7 + 5x^2)^{3/2}} dx = \int x \cdot (7 + 5x^2)^{-3/2} dx = \frac{1}{10} \int 10x \cdot (7 + 5x^2)^{-3/2} dx$$

$$= \frac{(7 + 5x^2)^{-1/2}}{-1/2} \cdot \frac{1}{10} + C = -\frac{2}{10 \sqrt{7 + 5x^2}} + C, C \in \mathbb{R}$$

$$r) \int \frac{x^3}{1 + x^8} dx = \frac{1}{4} \int \frac{4x^3}{1 + x^{4^2}} dx = \frac{1}{4} \arctg(x^4) + C, C \in \mathbb{R}$$

$$(2) a) \int \frac{e^{\arcsin x}}{\sqrt{1 - x^2}} dx = e^{\arcsin x} + C, C \in \mathbb{R}$$

$$b) \int \operatorname{tg}^2 x dx = \int \sec^2 x - 1 dx = \int \sec^2 x dx - \int 1 dx$$

$$= \operatorname{tg} n - n + c, \quad c \in \mathbb{R}$$

$$c) \int \frac{1}{n} \cos(\ln n) \, dn = \operatorname{sen}(\ln n) + c, \quad c \in \mathbb{R}$$

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$$a) \int n \sqrt{n+1} \, dn \longrightarrow \int t^2 - 1(t) \cdot 2t \, dt = \int (t^2 - 1) \cdot 2t^2 \, dt = \int 2t$$

$$t = \sqrt{n+1} \quad (\Rightarrow) \quad t^2 - 1 = n$$

$$dn = 2t \, dt$$

$$b) \int \frac{n}{1+\sqrt{n}} \, dn = \int \frac{t^2}{1+t} \cdot 2t \, dt = 2 \int \frac{t^3}{1+t} \, dt = 2 \int \frac{t^2}{\frac{1}{t} + 1} \, dt$$

$$t = \sqrt{n} \quad (\Rightarrow) \quad t^2 = n$$

$$2t \, dt = dn$$

$$\textcircled{9} \text{ a) } \int (n+1) \operatorname{sen} n = (n+1)(-\cos n) - \int -\cos n \, dn$$

$$\begin{array}{lll} u = n+1 & dv = \operatorname{sen} n & = -\cos(n)(n+1) - \operatorname{sen} n + C, C \in \mathbb{R} \\ du = 1 & v = -\cos n & \end{array}$$

$$\text{b) } \int n \cos(n) \, dn = n \operatorname{sen} n - \int \operatorname{sen}(n) \, dn = n \operatorname{sen} n + \cos n + C$$

$$\begin{array}{ll} u = n & dv = \cos n \\ du = 1 & v = \operatorname{sen} n \end{array}$$

$$\text{c) } \int n^2 \cos n \, dn = n^2 \operatorname{sen} n - \int 2n \operatorname{sen} n$$

$$\begin{array}{ll} u = n^2 & dv = \cos n \\ du = 2n & v = \operatorname{sen} n \end{array}$$

$$\begin{array}{ll} u = 2n & dv = \operatorname{sen} n \\ du = 2 & v = -\cos n \end{array}$$

$$= n^2 \operatorname{sen} n - \left(-2n \cos(n) - \int 2(-\cos n) \, dn \right) =$$

$$= x^2 \sin x + 2x \cos x + 2 \sin x + C =$$

$$d) \int e^{-3x} (2x+3) dx = (2x+3) \left(\frac{e^{-3x}}{3} \right) - \int \frac{e^{-3x}}{3} \cdot 2 dx$$

$$dv = e^{-3x} \quad v = 2x+3$$

$$v = \frac{e^{-3x}}{3} \quad dv = 2$$

$$= (2x+3) \frac{e^{-3x}}{3} - \frac{2}{3} \int e^{-3x} dx = (2x+3) \left(\frac{e^{-3x}}{3} \right) + \frac{2}{9} \int -3e^{-3x} dx$$

$$= (2x+3) \left(\frac{e^{-3x}}{3} \right) + \frac{2}{9} e^{-3x} + C$$

$$e) \int \ln^2 x dx = \int 1 \cdot \ln^2 x dx = x \ln^2 x - \int x \cdot 2 \ln x \cdot \frac{1}{x} dx$$

$$dv = 1$$

$$v = x$$

$$v = \ln^2 x$$

$$dv = 2 \ln x \cdot \frac{1}{x}$$

$$= x \ln^2(x) - 2 \int \ln x \, dx = x \ln^2(x) - 2 \left(x \ln x - \int \frac{1}{x} \cdot x \, dx \right)$$

$$\begin{aligned} u &= \ln x & dv &= 1 & & = x \ln^2 x - 2x \ln x + 2x + C \\ du &= \frac{1}{x} & v &= x \end{aligned}$$

$$g) \int \ln(x^2 + 1) \, dx = x \cdot \ln(x^2 + 1) - \int x \cdot \frac{2x}{x^2 + 1} \, dx =$$

$$\begin{aligned} u &= \ln(x^2 + 1) & dv &= 1 \\ du &= \frac{2x}{x^2 + 1} & v &= x \end{aligned}$$

$$= x \ln(x^2 + 1) - \int \frac{2x}{x^2 + 1} \cdot x \, dx = x \ln(x^2 + 1) - 2 \int \frac{x^2 + 1 - 1}{x^2 + 1} \, dx =$$

$$= x \ln(x^2 + 1) - 2 \int \frac{x^2 + 1}{x^2 + 1} - \frac{1}{x^2 + 1} \, dx = x \ln(x^2 + 1) - 2x + 2 \operatorname{arctg}(x) + C$$

$$h) \int n \arctg n = \frac{n^2}{2} \arctg n - \int \frac{n^2}{2} \cdot \frac{1}{1+n^2} dn =$$

$$v = \arctg n \quad dv = \frac{1}{1+n^2}$$

$$u = \frac{n^2}{2}$$

$$= \frac{n^2}{2} \arctg n - \frac{1}{2} \int \frac{n^2 + 1 - 1}{n^2 + 1} dn = \frac{n^2}{2} \arctg n - \frac{1}{2} \int 1 - \frac{1}{n^2 + 1} dn$$

$$= \frac{n^2}{2} \arctg n - \frac{n}{2} + \frac{\arctg n}{2} + c$$

$$\textcircled{10} a) \int n \sqrt{n+1} dn = \int (t^2 - 1) \cdot t \cdot 2t dt = \int (t^2 - 1) \cdot 2t^2 dt$$

$$t = \sqrt{n+1} \Leftrightarrow t^2 - 1 = n$$

$$2t dt = dn$$

$$= \int 2t^4 - 2t^2 dt = \frac{2t^5}{5} - \frac{2t^3}{3} + c$$

$$= \frac{2\sqrt{n+1}^5}{5} - 2\frac{\sqrt{n+1}^3}{3} + C, \quad C \in \mathbb{R}$$

$$b) \int \frac{u}{1+\sqrt{u}} du = \int \frac{(t-1)^2}{t} \cdot 2t^2 - 2t dt = 2 \int \frac{(t-1)^2}{t} \cdot t^2 - t dt$$

$$t = \sqrt{u} + 1 \quad = 2 \int (t-1)^2 \cdot t dt - 2 \int t dt = 2 \int (t-1)^2 \cdot t dt - t^2$$

$$u = (t-1)^2$$

$$du = 2(t-1) \cdot t = 2(t^2 - 2t - 1) \cdot t dt - t^2 =$$

$$= (2t-2) \cdot t$$

$$= 2t^2 - 2t dt = 2 \int t^3 - 2t^2 - t dt - t^2 = 2 \left(\frac{t^4}{4} - \frac{4t^3}{3} - \frac{2t^2}{2} - t^2 \right)$$

$$= \frac{t^4}{2} - \frac{4t^3}{3} - 2t^2 + C$$

$$= \frac{(\sqrt{n+1})^4}{2} - 4 \frac{(\sqrt{n+1})^3}{3} - 2(\sqrt{n+1})^2 + C, \quad C \in \mathbb{R}$$

$$c) \int \frac{1}{x^2 \sqrt{1-x^2}} dx = \int \frac{\cos(u)}{\sin^2(u) \cdot \sqrt{1-\sin^2(u)}} du = \int \frac{\cos(u)}{\sin^2(u) \cdot \sqrt{\cos^2(u)}} dx$$

$$x = \sin(u) \Rightarrow \arcsin(x) = u$$

$$dx = \cos(u) du$$

$$= \int \frac{1}{\sin^2(u)} du = ??$$

$$d) \frac{1}{x^2 \sqrt{x^2+4}} dx = \int \frac{1}{x^2 \sqrt{x^2+2^2}} dx =$$

$$x = 2 \operatorname{tg}(t) \Rightarrow \operatorname{arctg}\left(\frac{x}{2}\right) = t$$

$$dx = 2 \sec^2(t) dt$$

$$\begin{aligned}
&= \int \frac{2 \sec^2(t)}{2 \operatorname{tg}^2(t) \cdot \sqrt{4 \operatorname{tg}^2(t) + 4}} dt = \int \frac{\sec^2 t}{\operatorname{tg}^2(t) \cdot \sqrt{4 \sec^2(t)}} dt \\
&= \int \frac{\sec^2 t}{\operatorname{tg}^2(t) \cdot 2 \cdot \sqrt{\sec^2 t}} dt = \int \frac{\sec t}{\operatorname{tg}^2(t)} dt = \frac{1}{2} \int \frac{2 \sec(t)}{\operatorname{tg}^2(t)} dt
\end{aligned}$$

???

Copiando da aula)

$$j) \frac{1}{x^2 \sqrt{9 - x^2}} dx = \int \frac{1}{9 \operatorname{sen}^2 t \sqrt{9 - 9 \operatorname{sen}^2 t}} \cdot 3 \cos t$$

$$x = 3 \operatorname{sen}(t) \Rightarrow \arcsen\left(\frac{x}{3}\right) = t$$

$$dx = 3 \cos(t)$$

$$= \int \frac{3 \cos t}{9 \sin^2 t \sqrt{9(1 - \sin^2 t)}} dt = \int \frac{3 \cos t}{9 \sin^2 t \cdot 3 \sqrt{\cos^2 t}} dt =$$

$$= \frac{3 \cos t}{9 \sin^2 t \cdot 3 \cos t} dt = \int \frac{1}{9} \operatorname{cosec}^2 t dt = -\frac{1}{9} \cotg(t) + C$$

$$= -\frac{1}{9} \sqrt{\operatorname{cosec}^2 t - 1} + C =$$

/ Ficha 2 - Parte 2

$$2- a) F(x) = \int_0^{x^2} e^{t^2} dt$$

$$F'(x) = ?$$

$$F'(x) = e^{(x^2)^2} \cdot (x^2)' - e^{0^2} \cdot (0)' = e^{x^4} \cdot 2x$$

↙ T. fundamental
do cálculo

$$b) F(x) = \int_0^x \frac{t^2}{t^2+1} dt$$

$$F'(x) = \frac{x^2}{x^2+1} \cdot 1 - 0 \cdot 0 = \frac{x^2}{x^2+1}$$

$$c) F(x) = \int_x^0 e^{-s^2} ds$$

$$F'(x) = e^{-0^2} \cdot 0 - e^{-x^2} \cdot 1 = -e^{-x^2}$$

$$d) F(x) = \int_1^x (\sin t^2 + e^{-t^2}) dt$$

$$F'(x) = (\sin x^2 + e^{-x^2}) \cdot 1 - \sin 1 + e^{-1^2} \cdot 0$$

$$= \sin x^2 + e^{-x^2}$$

$$4) F(x) = \int_0^{x^2} \sin t^2 dt$$

$$F'(\sqrt[4]{\frac{\pi}{4}}) = ?$$

	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
sen	1	$\frac{\sqrt{2}}{2}$	$\frac{3}{4}$
cos	$\frac{3}{4}$	$\frac{2}{2}$	1

$$f'(x) = \sin(x^4) \cdot (x^4)' = 2x \cdot \sin(x^4)$$

tan

$$\begin{aligned} f'\left(\sqrt[4]{\frac{\pi}{4}}\right) &= 2 \cdot \sqrt[4]{\frac{\pi}{4}} \cdot \sin\left(\frac{\pi}{4}\right) = 2 \sqrt[4]{\frac{\pi}{4}} \cdot \frac{\sqrt{2}}{2} = \sqrt[4]{\frac{\pi}{4}} \cdot \sqrt{2} \\ &= \sqrt[4]{\frac{\pi}{4}} \cdot \sqrt[4]{2^2} = \sqrt[4]{\frac{4\pi}{4}} = \sqrt[4]{\pi} \end{aligned}$$

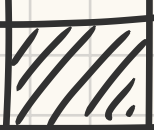
6) $G = \int_0^x e^{3t^4 + 4t^3} dt$

a) monotonic?

$$G'(x) = e^{3x^4 + 4x^3} \cdot 1 = e^{3x^4 + 4x^3}$$

!!

$$G'(x) = 0 \Rightarrow \text{imp.}$$

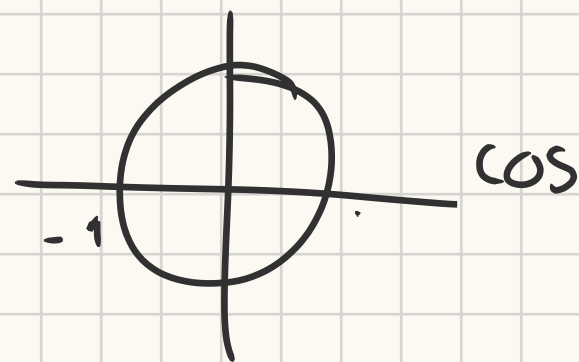
		0	
$G'(x)$			
$G(x)$			

$$8) \lim_{x \rightarrow 1} \frac{\int_1^x t \cos(1 - e^{1-t}) dt}{x^2 - 1} = \frac{2x \cdot x^2 \cos(1 - e^{1-x^2}) - \cos(1 - e^0) \cdot 0}{2x}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 \cdot 2x \cdot \cos(1 - e^{1-x^2})}{2x} = \lim_{x \rightarrow 1} x^2 \cos(1 - e^{1-x^2})$$

$$= 1 \cdot \cos(1 - e^0) = 1 \cdot \cos(0) = 1$$

1) São integráveis?



$$a) f: [0, 4] \rightarrow \mathbb{R} \quad f(x) = \cos(\underbrace{x^2 - 2x}_{0 \leq \pi})$$

f é contínua em $[0, 4]$ logo é integrável em $[0, 4]$

$$b) f: [-2, 1] \rightarrow \mathbb{R} \quad f(x) = \begin{cases} x+1 & , x \in [-2, 0[\\ 2 & , x = 0 \\ x & , x \in]0, 1] \end{cases}$$

limitada em $[-2, 1]$ ✓

$$\lim_{x \rightarrow 0^-} x+1 = 0$$

$$f(0) = 2$$

$$\lim_{x \rightarrow 0} 2 = 2$$

Só n é contínua em $f(0)$, logo
é integrável em $[-2, 1]$

$$\lim_{x \rightarrow 0^+} x = 0$$

$$12) \quad a) \int_0^2 6x^4 dx = \left[6 \frac{x^5}{5} \right]_0^2 = 6 \cdot \frac{2^5}{5} - 6 \cdot 0 = 6 \cdot \frac{32}{5} = \frac{192}{5}$$

$$b) \int_3^2 \frac{t^2}{3} - \sqrt{t} \, dt = \int_3^2 \frac{t^2}{3} \, dt - \int_3^2 t^{1/2} \, dt = \left[\frac{1}{3} \frac{t^3}{3} \right]_3^2 - \left[\frac{t^{3/2}}{3/2} \right]_3^2$$

$$= \left(\frac{2^3}{9} - \frac{3^3}{9} \right) - \left(\frac{2\sqrt{2^3}}{3} - \frac{2\sqrt{3^3}}{3} \right) = \frac{8}{9} - \frac{9}{9} - \left(\frac{2}{3} (\sqrt{2^3} - \sqrt{3^3}) \right)$$

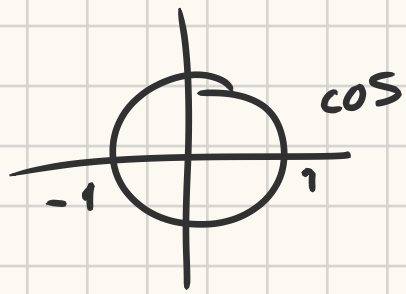
$$= -\frac{1}{9} - \frac{2}{3} (2\sqrt{2} - 3\sqrt{3}) = -\frac{1}{9} - \frac{4\sqrt{2}}{3} + 2\sqrt{3}$$

$$c) \int_{-4}^{-3} \frac{e^x}{3} \, dx = \left[\frac{1}{3} e^x \right]_{-4}^{-3} = \frac{1}{3} (e^{-3} - e^{-4}) = \frac{1}{3} e^3 (1 - e^{-1}) = \frac{e^3}{3} \left(1 - \frac{1}{e} \right)$$

$$d) \int_1^3 \frac{x^3}{\sqrt{x}} \, dx = \int_1^3 x^3 \cdot x^{-1/2} \, dx = \int_1^3 x^{\frac{6}{2} - \frac{1}{2}} \, dx = \int_1^3 x^{5/2} \, dx = \left[\frac{x^{7/2}}{7/2} \right]_1^3$$

$$= \frac{3^{7/2}}{7/2} - \frac{1^{7/2}}{7/2} = \frac{2\sqrt{3^7}}{7} - \frac{2}{7}$$

$$e) \int_0^1 \frac{1}{1+t^2} dt = \arctg(1) - \arctg(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$



$$i) \int_{-\pi}^0 \sin(3x) dx = \left[-\frac{1}{3} \cos(3x) \right]_{-\pi}^0 = -\frac{1}{3} (\cos(0) - \cos(\pi))$$

$$= -\frac{1}{3} (1 + 1) = -\frac{2}{3}$$

$$h) \int_e^{e^2} \frac{1}{x(\ln x)^2} dx = \int_e^{e^2} \frac{1}{x} \cdot (\ln x)^{-2} dx = \left[\frac{(\ln x)^{-1}}{-1} \right]_e^{e^2} = -\frac{1}{\ln e^2} + \frac{1}{\ln e}$$

$$= -\frac{1}{2} + \frac{1}{1} = \frac{1}{2}$$

$$p) \int_1^2 \frac{1}{x^2 + 2x + 5} dx = *$$

$$\begin{aligned} \text{CA} \quad x^2 + 2x + 5 &= x^2 + 2x + 1 + 4 = (x+1)^2 + 4 = 4 \left(\left(\frac{x+1}{2} \right)^2 + 1 \right) = \\ &= 4 \left(\left(\frac{x+1}{2} \right)^2 + 1 \right) \end{aligned}$$

$$= \int_1^2 \frac{1}{4 \left(\left(\frac{x+1}{2} \right)^2 + 1 \right)} dx = \frac{1}{4} \int_1^2 \frac{1}{\left(\frac{x+1}{2} \right)^2 + 1} = \frac{1}{2} \int_1^2 \frac{1/2}{1 + \left(\frac{x+1}{2} \right)^2} dx = \frac{1}{2} \left[\arctan \left(\frac{x+1}{2} \right) \right]_1^2$$

$$= \frac{1}{2} \left(\arctan \left(\frac{3}{2} \right) - \arctan \left(\frac{2}{2} \right) \right) = \frac{1}{2} \arctan \left(\frac{3}{2} \right) - \frac{\pi}{4}$$

$$\begin{aligned} 13 \quad a) \int_{-\ln 2}^{\ln 2} \frac{1}{e^x + 4} dx & \quad t = e^x + 4 \Rightarrow e^x = t - 4 \Rightarrow x = \ln(t - 4) \\ & \quad (\Rightarrow) dx = \frac{1}{t - 4} dt \end{aligned}$$

x	t
$-\ln 2$	$e^{-\ln 2} + 4 = e^{\ln 2^{-1}} + 4 = \frac{1}{2} + 4 = \frac{1}{2} + \frac{8}{2} = \frac{9}{2}$
$\ln 2$	$e^{\ln 2} + 4 = 6$

$$= \int_{\frac{9}{2}}^6 \frac{1}{t} \cdot \frac{1}{t-4} dt = \int_{\frac{9}{2}}^6 \frac{1}{(t-4)t} dt = *$$

CA

$$\frac{1}{(t-4)t} = \frac{A}{t-4} + \frac{B}{t} = \frac{A(1) + B(t-4)}{(t-4)t}$$

$$\begin{cases} A + B = 0 \\ -4B = 1 \end{cases} \quad \begin{cases} A = \frac{1}{4} \\ B = -\frac{1}{4} \end{cases}$$

$$= * \int_{\frac{9}{2}}^6 \frac{\frac{1}{4}}{t-4} + \frac{-\frac{1}{4}}{t} dt = \frac{1}{4} \int \frac{1}{t-4} - \frac{1}{t} dt = \frac{1}{4} \left(\ln |t-4| - \ln |t| \right) \Big|_{\frac{9}{2}}^6$$

$$= \frac{1}{4} \left((\ln(2) - \ln(6)) - (\ln(\frac{1}{2}) - \ln(\frac{9}{2})) \right) = \frac{1}{4} \left(\ln\left(\frac{2}{6}\right) - \ln\left(\frac{\frac{1}{2}}{\frac{9}{2}}\right) \right)$$

$$= \frac{1}{4} \ln \left(\frac{1/3}{1/4} \right) = \frac{1}{4} \ln \left(\frac{4}{3} \right) = \frac{1}{4} \ln \left(\frac{3}{4} \right)$$

$$b) \int_0^1 \frac{x}{1+x^4} dx = \frac{1}{2} \int_0^1 \frac{2x}{1+(x^2)^2} = \frac{1}{2} \left[\arctg(x^2) \right]_0^1 =$$

$$\frac{\sin \frac{1}{2}}{\cos \frac{1}{2}}$$



$$= \frac{1}{2} \left(\arctg(1) - \arctg(0) \right) = \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{8}$$

$$c) \int_0^1 \sqrt{4-x^2} dx \xrightarrow{\substack{x=2\sin(t) \Rightarrow t=\arcsin(\frac{x}{2}) \\ dx=2\cos(t) dt}} \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2(t)} \cdot 2\cos(t) dt$$

x	t
1	$\arcsin(\frac{1}{2}) = \frac{\pi}{6}$
0	$\arcsin(0) = 0$

$$4-4\sin^2 = 4\cos^2$$

$$= \int_0^{\pi/6} 2\cos t \cdot \sqrt{4\cos^2(t)} dt = \int_0^{\pi/6} 2\cos t \cdot 2\cos t dt = 4 \int_0^{\pi/6} \cos^2 t dt$$

$$= 4 \int_0^{\pi/6} \frac{1 + \cos(2t)}{2} dt = 2 \int_0^{\pi/6} 1 + \cos(2t) dt$$

$$= 2 \int_0^{\pi/6} 1 dt + \int_0^{\pi/6} 2 \cos(2t) dt = \left[2t \right]_0^{\pi/6} + \left[\sin(2t) \right]_0^{\pi/6} =$$

$$= 2 \frac{\pi}{6} - 0 + \sin\left(\frac{2\pi}{6}\right) = \frac{\pi}{3} + \frac{\sqrt{3}}{2} + C, \quad C \in \mathbb{R}$$

$$d) \int_1^e x \ln x \, dx = \ln x \cdot \frac{x^2}{2} \Big|_1^e - \int_1^e \frac{x^2}{2} \cdot \frac{1}{x} dx =$$

$v = \ln x \quad dv = \frac{1}{x}$
 $du = x \quad u = \frac{x^2}{2}$

$$= \frac{x^2}{2} \ln x \Big|_1^e - \int_1^e \frac{x^2}{2x} dx = \frac{e^2}{2} \cdot 1 - \frac{1}{2} \cdot 0 - \frac{1}{2} \int_1^e x \, dx$$

$$= \frac{e^2}{2} - \frac{1}{2} - \frac{1}{2} \left[\frac{x^2}{2} \right]_1^e = \frac{e^2}{2} - \frac{1}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) = \frac{e^2}{2} - \frac{1}{2} - \frac{e^2}{4} + \frac{1}{4}$$

$$= \frac{2e^2 - e^2}{4} - \frac{2}{4} + \frac{1}{4} = \frac{e^2}{4} - \frac{1}{4} = \frac{e^2 - 1}{4}$$

$$e) \int_1^e \ln^2 x \, dx = \int_1^e 1 \cdot \ln^2 x \, dx = \ln^2(x) \cdot x \Big|_1^e - \int_1^e 2 \cdot \ln x \cdot \frac{1}{x}$$

$$u = \ln^2 x$$

$$dv = 1$$

$$du = 2 \ln x \cdot \frac{1}{x}$$

$$v = x$$

$$= \ln^2(e) \cdot e - \ln^2(1) - 1 - 2 \int_1^e \frac{1}{x} \cdot \ln(x) \, dx = e - 0 - 1 - 2 \left[\frac{\ln^2(x)}{2} \right]_1^e$$

$$= e - 1 - 2 \left(\frac{\ln^2(e)}{2} - \frac{\ln^2(1)}{2} \right) = e - 1 - 2 \left(\frac{1}{2} - \frac{0}{2} \right) = e - 1 - \frac{2}{2} = e - 2 + c$$

$$a) \int_0^2 f(x) dx, \quad f(x) = \begin{cases} 2 & \text{se } 0 \leq x < 1 \\ \frac{1}{x} & \text{se } 1 \leq x \leq 2 \end{cases}$$

$$\int_0^2 f(x) = \int_0^1 f(x) + \int_1^2 f(x) =$$

$$= \int_0^1 2 dx + \int_1^2 \frac{1}{x} dx = [2x]_0^1 + [\ln x]_1^2$$

$$= 2 \cdot 1 + \ln(2) - \ln(1) = 2 + \ln(2) + C, \quad C \in \mathbb{R}$$

$$b) \int_{-1}^1 f(x) \quad , \quad f(x) = \begin{cases} \frac{2}{1+x^2} & , \quad x \in [-1, 0[\\ 7 & , \quad x = 0 \\ \frac{1}{1+x} & , \quad x \in]0, 1] \end{cases}$$

$$\int_{-1}^1 f(x) = \int_{-1}^0 \frac{2}{1+x^2} + \int_0^1 \frac{1}{1+x}$$

$$= 2 \left[\arctg(x) \right]_{-1}^0 + \left[\ln(1+x) \right]_0^1$$

$$= 2(\arctg(0) - \arctg(-1)) + \ln(2) - \ln(1)$$

$$= 2\left(0 - \frac{\pi}{4}\right) + \ln(2) - 0 = \frac{\pi}{2} + \ln(2)$$

$$c) \int_{-1}^3 f(x), \quad f(x) = \begin{cases} \frac{x}{1+x^2}, & x \neq 1 \\ 5, & x = 1 \end{cases}$$

$$\int_{-1}^3 f(x) = \int_{-1}^1 \frac{x}{1+x^2} + \int_1^3 \frac{x}{1+x^2} = *$$

C.A

$$\int \frac{x}{1+x^2} = \int \frac{\frac{\sqrt{u-1}}{u}}{\frac{1}{2\sqrt{u-1}}} du = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln(u)$$

$$u = x^2 + 1 \Rightarrow \sqrt{u-1} = x \qquad = \frac{1}{2} \ln(x^2 + 1) + C$$

$$\frac{1}{2} (u-1)^{-1/2} = dx$$

$$\Rightarrow \frac{1}{2\sqrt{u-1}} du = dx$$

$$* \left[\frac{\ln(x^2 + 1)}{2} \right]_{-1}^1 + \left[\frac{\ln(x^2 + 1)}{2} \right]_1^3 =$$

$$= \frac{1}{2} \left(\ln(2) - \ln(2) + \ln(10) - \ln(2) \right) = \frac{1}{2} \ln\left(\frac{10}{2}\right) = \frac{1}{2} \ln(5)$$

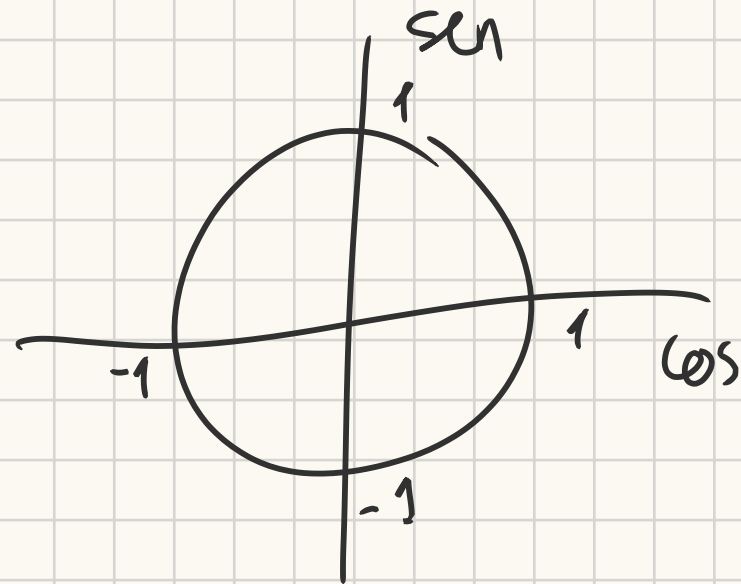
$$d) \int_0^{2\pi} f(x) dx, \quad f(x) = \begin{cases} -2, & x \in [0, \frac{\pi}{2}] \\ \cos x, & x \in [\frac{\pi}{2}, \frac{3\pi}{2}] \\ \sin x, & x \in [\frac{3\pi}{2}, 2\pi] \end{cases}$$

$$\int_0^{2\pi} f(x) = \int_0^{\pi/2} 2 + \int_{\pi/2}^{3\pi/2} \cos x + \int_{3\pi/2}^{2\pi} \sin x$$

$$= [2x]_0^{\pi/2} + [\sin x]_{\pi/2}^{3\pi/2} + [-\cos x]_{3\pi/2}^{2\pi}$$

$$= 2 \frac{\pi}{2} + \sin\left(\frac{3\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right) - \cos(2\pi) + \cos\left(\frac{3\pi}{2}\right)$$

$$= \pi - 1 - 1 - 1 + 0 = \pi - 3$$



14 (De novo)

$$a) \int_0^2 f(x) \quad , \quad f(x) = \begin{cases} 2 & \text{se } 0 \leq x < 1 \\ \frac{1}{x} & \text{se } 1 \leq x \leq 2 \end{cases}$$

$$\int_0^2 f(x) = \int_0^1 2 \, dx + \int_1^2 \frac{1}{x} \, dx = \left[2x \right]_0^1 + \left[\ln(x) \right]_1^2 =$$

$$= 2 \cdot 1 - 2 \cdot 0 + \ln(2) - \ln(1) = 2 + \ln(2)$$

$$b) \int_{-1}^1 f(x) \quad , \quad f(x) = \begin{cases} \frac{2}{1+x^2} & , \quad x \in [-1, 0] \\ 7 & , \quad x = 0 \\ \frac{1}{1+x} & , \quad x \in]0, 1] \end{cases}$$

$$\int_{-1}^1 f(x) = \int_{-1}^0 \frac{2}{1+x^2} \, dx + \int_0^1 \frac{1}{1+x} \, dx$$

$$= 2 \left[\arctg(x) \right]_{-1}^0 + \left[\ln(1+x) \right]_0^1 = 2(\arctg(0) - \arctg(-1)) + (\ln(2) - \ln(1))$$

$$= 2\left(0 + \frac{\pi}{4}\right) + \ln(2) - 0 = 2\frac{\pi}{4} + \ln(2) = \frac{\pi}{2} + \ln(2)$$

sen

cos

tan $\frac{\sqrt{3}}{3}$ 1 $\sqrt{3}$

$$c) \int_{-1}^3 f(x) \quad , \quad f(x) = \begin{cases} \frac{x}{1+x^2} & , \quad x \neq 1 \\ 5 & , \quad x = 1 \end{cases}$$

$$\int_{-1}^3 f(x) = \int_{-1}^1 \frac{x}{1+x^2} + \int_1^3 \frac{x}{1+x^2} = *$$

$$CA \quad \int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{2x}{1+x^2} = \frac{1}{2} \ln(1+x^2)$$

$$* \frac{1}{2} \left(\left[\ln(1+x^2) \right]_{-1}^1 + \left[\ln(1+x^2) \right]_1^3 \right) =$$

$$= \frac{1}{2} \left(\ln(2) - \ln(2) + \ln(10) - \ln(2) \right) = \frac{1}{2} \left(\ln(10) - \ln(2) \right) = \frac{1}{2} \left(\ln\left(\frac{10}{2}\right) \right)$$

$$= \frac{1}{2} \ln(5) + C, \quad C \in \mathbb{R}$$

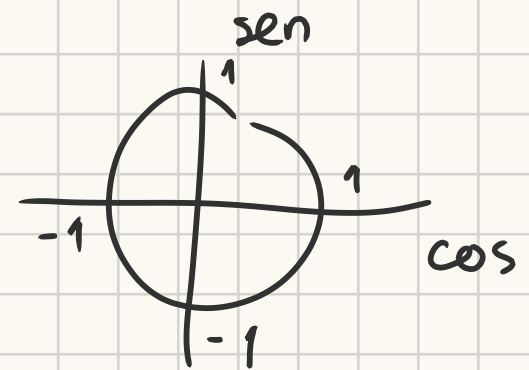
$$d) \int_0^{2\pi} f(x) \quad , \quad f(x) = \begin{cases} -2 & , \quad x \in \left[0, \frac{\pi}{2}\right] \\ \cos x & , \quad x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \\ \sin x & , \quad x \in \left]\frac{3\pi}{2}, 2\pi\right] \end{cases}$$

$$\int_0^{2\pi} f(x) dx = \int_0^{\pi/2} -2 dx + \int_{\pi/2}^{\frac{3\pi}{2}} \cos x dx + \int_{\frac{3\pi}{2}}^{2\pi} \sin x dx$$

$$= \left[-2x\right]_0^{\pi/2} + \left[\sin x\right]_{\pi/2}^{\frac{3\pi}{2}} + \left[-\cos x\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= -2\frac{\pi}{2} - 0 + \sin \frac{3\pi}{2} - \sin \frac{\pi}{2} + (-\cos 2\pi) + \cos\left(\frac{3\pi}{2}\right)$$

$$= -\pi - 1 - 1 - 1 + 0 = -\pi - 3$$



$$(15) \quad f(x) = \frac{1}{x \ln(x)}$$

a) primitiva que se anula em $x = e^2$?

$$\int \frac{1}{x \ln(x)} dx = \int \frac{1}{x} \cdot \frac{1}{\ln(x)} dx = \int \frac{\overbrace{\frac{1}{x}}^{u'}}{\underbrace{\ln(x)}_u} dx = \ln |\ln(x)| + C$$

$$\ln |\ln(e^2)| + C = 0 \Leftrightarrow \ln |2| + C = 0 \Leftrightarrow \ln(2) = -C \Leftrightarrow -\ln(2) = C$$

$$\therefore \ln |\ln(x)| - \ln(2)$$

b) Área entre $x = e$, $x = e^3$, eixo de abscissas e f ?

$$f(x) = \frac{1}{x \ln(x)} > 0$$

$$\int_e^{e^3} \frac{1}{x \ln(x)} \quad , \text{ pela a linha anterior } \rightarrow \left[\ln |\ln |x|| \right]_e^{e^3}$$

$$\ln(\ln(e^3)) - \ln(\ln(e)) = \ln(3) - \ln(1) = \ln\left(\frac{3}{1}\right) = \ln(3)$$

(16) Área entre $x=0$, $x=2$, eixo das abscissas e por $g(x)$

$$g(x) = x \ln(x+1) \geq 0$$

$$\int_0^2 x \ln(x+1) dx = \ln(x+1) \cdot \frac{x^2}{2} \Big|_0^2 - \int_0^2 \frac{x^2}{2} \cdot \frac{1}{x+1} dx$$

$$u = \ln(x+1) \quad dv = x$$

$$du = \frac{1}{x+1}$$

$$v = \frac{x^2}{2}$$

$$= \ln(x+1) \cdot \frac{x^2}{2} \Big|_0^2 - \frac{1}{2} \int_0^2 \frac{x^2}{x+1} dx = \ln(x+1) \cdot \frac{x^2}{2} \Big|_0^2 - \frac{1}{2} \int_0^2 x^{-1} + \frac{1}{x+1}$$

$$\frac{x^2}{-x^2-x} \cdot \frac{x+1}{x-1} = \ln(x+1) \frac{x^2}{2} \Big|_0^2 - \frac{1}{2} \left(\int_0^2 x^{-1} + \int_0^2 \frac{1}{x+1} \right)$$

$$\frac{-x}{+x+1} + 1$$

$$= \ln(3) \frac{4}{2} - \ln(1) \cdot \frac{1}{2} - \frac{1}{2} \left(\left[\frac{x^2}{2} - x \right]_0^2 + \left[\ln(x+1) \right]_0^2 \right) =$$

$$= 2\ln(3) - 0 - \frac{1}{2} \left(\frac{4}{2} - 2 - 0 + \ln(3) - \ln(1) \right) =$$

$$= 2\ln(3) - \frac{1}{2} \left(\ln(3) \right) = 2\ln(3) - \frac{\ln(3)}{2} = \frac{1}{2} \left(4\ln(3) - \ln(3) \right) = \frac{3}{2} \ln(3)$$

treino de divisões

$$\begin{array}{r|l} x^3 + 0x^2 + px + q & x^2 + 2x + 5 \\ -x^3 - 2x^2 - 5x & \\ \hline -2x^2 + (p-5)x + q & \end{array} \quad \begin{array}{l} \\ \\ x-2 \end{array}$$

$$\frac{+2n^2 + 4n + 10}{(p-1)n + q10}$$

$$\begin{array}{r|l} n^3 - 3n^2 + 5n - 3 & n - 1 \\ - n^3 + n^2 & \hline - 2n^2 + 5n - 3 & \\ + 2n^2 - 2n & \hline 3n - 3 & \\ - 3n + 3 & \hline 0 & \end{array}$$

$$\begin{array}{r|l} 6n^4 - n^3 + 3n^2 - n + 1 & 2n^2 + n - 3 \\ - 6n^4 - 3n^3 + 9n^2 & \hline - 4n^3 + 12n^2 - n + 1 & \\ 4n^3 + 2n^2 - 6n & \hline 14n^2 - 7n + 1 & \\ - 14n^2 - 7n + 21 & \hline - 14n + 22 & \end{array}$$

$$\frac{6n^4 - n^3 + 3n^2 - n + 1}{2n^2 + n - 3} = 3n^2 - 2n + 7 + \frac{-14n + 22}{2n^2 + n - 3}$$

$$\frac{-2n^2}{n} = -2n$$

$$\frac{3n}{n} = 3$$

$$\frac{n^3 - 3n^2 + 5n - 3}{n - 1} = n^2 - 2n + 3$$

$$\frac{6n^4}{2n^2} = 3n^2$$

$$\frac{-4n^3}{2n^2} = -2n$$

$$\frac{14n^2}{2n^2} = 7$$

$$\begin{array}{r} n^2 + n - 2 \quad | \quad n - 1 \\ -n^2 + n \\ \hline 2n - 2 \\ -2n + 2 \\ \hline 0 \end{array}$$

$$\frac{n^2}{n} = n$$

$$\frac{2n}{n} = 2$$

$$\frac{n^2 + n - 2}{n - 1} = n + 2$$

(17) Area entre $n=0$, $n=2$, eixo das abscissas e $g(n) = \frac{e^{2n} + 1}{e^n + 1}$

$$\frac{e^{2n} + 1}{e^n + 1} > 0$$

$$\int_0^2 \frac{e^{2n} + 1}{e^n + 1} dn$$

n	t
0	$e^0 + 1 = 2$
2	$e^2 + 1$

$$t = e^n + 1 \Leftrightarrow \ln(t-1) = n \Leftrightarrow t-1 = e^n$$

$$\frac{1}{t-1} dt = dn$$

$$\int_2^{e^2+1} \frac{(t-1)^2 + 1}{t(t-1)} \cdot dt$$

$$= \int_2^{e^2+1} \frac{t^2 - 2t + 1 + 1}{t^2 - t} dt = \int_2^{e^2+1} \frac{t^2 - 2t + 2}{t^2 - t} dt = \int_2^{e^2+1} 1 - \frac{t-2}{t^2-t}$$

$$\frac{t^2 - 2t + 2}{-t^2 + t} \quad \left| \frac{t^2 - t}{1} \right.$$

$$= \int_2^{e^2+1} 1 dt - \int_2^{e^2+1} \frac{t-2}{t(t-1)}$$

$$= \left[t \right]_2^{e^2+1} - \int_2^{e^2+1} \frac{t-2}{t(t-1)} dt \quad *$$

CA

$$\frac{t-2}{t(t-1)} = \frac{A}{t} + \frac{Bt+C}{t-1} = \frac{A(t-1) + B(t^2) + C(t)}{t(t-1)} =$$

$$= \frac{B(t^2) + A + C(t) + A(-1)}{t(t-1)}$$

$$\begin{cases} B=0 \\ A+C=1 \\ A=2 \end{cases}$$

$$\begin{cases} B=0 \\ C=1-2 \\ A=2 \end{cases}$$

$$\begin{cases} B=0 \\ C=-1 \\ A=2 \end{cases}$$

$$\rightarrow \frac{2}{t} + \frac{-1}{t-1}$$

$$* \left[t \right]_2^{e^2+1} - \int_2^{e^2+1} \frac{2}{t} dt - \int_2^{e^2+1} \frac{1}{t-1} dt =$$

$$= \left[t \right]_2^{e^2+1} - \left[2 \ln |t| \right]_2^{e^2+1} - \left[\ln |t-1| \right]_2^{e^2+1} =$$

$$= \left[e^x + 1 \right]_0^2 - \left[2 \ln(e^x + 1) \right]_0^2 - \left[\ln |e^x| \right]_0^2 =$$

$$= e^2 + 1 - (e^0 + 1) - 2 \left(\ln(e^2 + 1) - \ln(e^0 + 1) \right) - (2 - 0)$$

$$= e^2 + 1 - 1 - 1 - 2 \ln \left(\frac{e^2 + 1}{2} \right) - 2 = e^2 - 1 - 2 - 2 \ln \left(\frac{e^2 + 1}{2} \right)$$

$$= e^2 - 3 - 2 \ln \left(\frac{e^2 + 1}{2} \right)$$

18) $f(x) = x^3 - 3x^2 + 2x$. Anea entre $f(x)$, $x=0$, $x=2$ e O_x ?

$$x^3 - 3x^2 + 2x = 0 \Leftrightarrow (x-1)(x^2 - 2x) = 0 \Leftrightarrow$$

$$\Leftrightarrow (x-1) = 0 \vee (x^2 - 2x) = 0$$

$$\Leftrightarrow x = 1 \vee x^2 - 2x + 0 = 0$$

$$\Leftrightarrow \underline{x=1} \vee \underline{x=0} \vee \underline{x=2}$$

$$\begin{array}{r|rrrr} & 1 & -3 & 2 & \\ 1 & 1 & -2 & 0 & \end{array}$$

CA $x^2 - 2x + 0 = 0$

$$x = \frac{2 \pm \sqrt{4 - 4(1 \cdot 0)}}{2} \Leftrightarrow x = \frac{2 \pm \sqrt{4}}{2} \Leftrightarrow x = \frac{2 \pm 2}{2}$$

$$\Leftrightarrow x = \frac{2-2}{2} \vee x = \frac{2+2}{2} \Leftrightarrow x = 0 \vee x = 2$$

$$\text{em } n = -1 \quad (-1)^3 - 3(-1)^2 + 2(-1) = -1 - 3 - 2 < 0$$

$$\begin{aligned} \text{em } n = \frac{1}{2} \quad \left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right) &= \frac{1}{8} - \frac{3}{4} + \frac{2}{2} = \\ &= \frac{1}{8} - \frac{6}{8} + \frac{8}{8} = \frac{3}{8} > 0 \end{aligned}$$

$$\text{em } n = \frac{3}{2} \quad \left(\frac{3}{2}\right)^3 - 3\left(\frac{3}{2}\right)^2 + 2\left(\frac{3}{2}\right) = \frac{27}{8} - \frac{27}{4} + 3 =$$

$$= \frac{27}{8} - \frac{54}{8} + \frac{24}{8} = -\frac{3}{8} < 0$$

$$\int_0^2 f(x) = \int_0^1 f(x) - \int_1^2 f(x) =$$

$$= \int_0^1 x^3 - 3x^2 + 2x - \int_1^2 x^3 - 3x^2 + 2x$$

$$= \left[\frac{x^4}{4} - \cancel{3} \frac{x^3}{\cancel{3}} + \cancel{2} \frac{x^2}{\cancel{2}} \right]_0^1 - \left[\frac{x^4}{4} - x^3 + x^2 \right]_1^2$$

$$= \frac{1^4}{4} - 1^3 + 1^2 - 0 - \left(\frac{2^4}{4} - 2^3 + 2^2 - \left(\frac{1^4}{4} - 1 + 1 \right) \right)$$

$$= \frac{1}{4} - \frac{16}{4} + 8 - 4 + \frac{1}{4} = -\frac{14}{4} + 4 = -\frac{14}{4} + \frac{16}{4} = \frac{2}{4} = \frac{1}{2}$$

(19) Anea este $f(x) = \frac{4 + \sin^2 x}{1 + 4x^2}$, $g(x) = \frac{\sin^2 x}{1 + 4x^2}$, $x = 0$, $x = \frac{1}{2}$
e 0

$$\frac{4 + \sin^2 x}{1 + 4x^2} > 0, \quad \frac{\sin^2 x}{1 + 4x^2} \geq 0, \quad \frac{4 + \sin^2 x}{1 + 4x^2} > \frac{\sin^2(x)}{1 + 4x^2}$$

$$\begin{aligned}
 & \int_0^{1/2} \frac{4 + \sin^2 u}{1 + 4u^2} - \frac{\sin^2 u}{1 + 4u^2} du = \int_0^{1/2} \frac{4 + \sin^2 u - \sin^2 u}{1 + 4u^2} du \\
 & = \int_0^{1/2} \frac{4}{1 + 4u^2} du = 2 \int_0^{1/2} \frac{2}{1 + (2u)^2} du = 2 \left[\arctg(2u) \right]_0^{1/2} \\
 & = 2 \left(\arctg(1) - \arctg(0) \right) = 2 \left(\frac{\pi}{4} - 0 \right) = 2 \frac{\pi}{4} = \frac{\pi}{2}
 \end{aligned}$$

(20) Area entre $f(u) = u^2$ e $g(u) = u$?

$$u^2 = u \Leftrightarrow u^2 - u = 0 \Leftrightarrow u(u - 1) = 0 \Leftrightarrow u = 0 \vee u = 1$$

$$u^2 > u$$

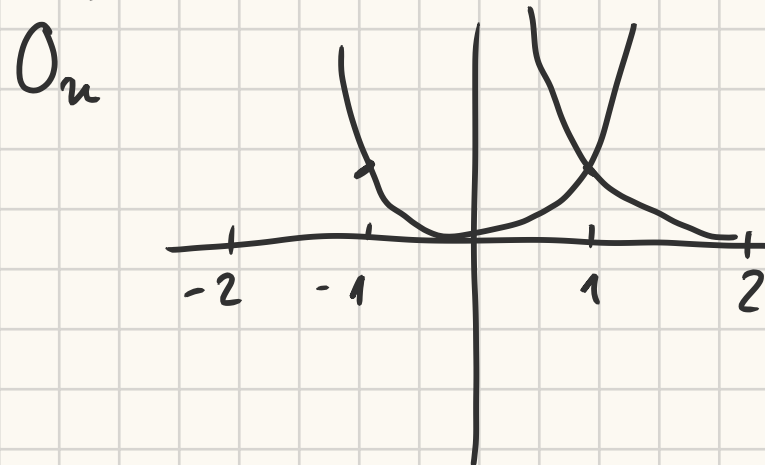
em $x = \frac{1}{2}$) $\left(\frac{1}{2}\right)^2 = \frac{1}{4} < \frac{1}{2}$

$$\int_0^1 x - x^2 dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$$

(21) Area entre $f(x) = \frac{1}{x}$, $g(x) = x^2$, $x=2$ e $y=0$,

$$\frac{1}{x} = x^2 \Leftrightarrow 1 = x^3 \Leftrightarrow x = 1$$

1 $\rightarrow$ 2



$$\int_0^1 x^2 + \int_1^2 \frac{1}{x} = \left[\frac{x^3}{3} \right]_0^1 + \left[\ln|x| \right]_1^2 = \frac{1}{3} + \ln(2) - \ln(1)$$

$$= \frac{1}{3} + \ln(2)$$

21) Area entre $f(x) = e^{2x+1}$, $g(x) = xe^{2x+1}$, $x = -1$ e $x = -\frac{1}{2}$

$$e^{2x+1} = xe^{2x+1} \Leftrightarrow x = 1$$

$$e^{-2+1} = e^{-1} = \frac{1}{e}$$

$$-1e^{-2+1} = -1e^{-1} = -\frac{1}{e} \rightarrow$$

em $x < 0$) $f(x) > g(x)$

em $x = -\frac{2}{3}$)

$$f(-\frac{2}{3}) = e^{-\frac{4}{3} + \frac{3}{3}} = e^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{e}}$$

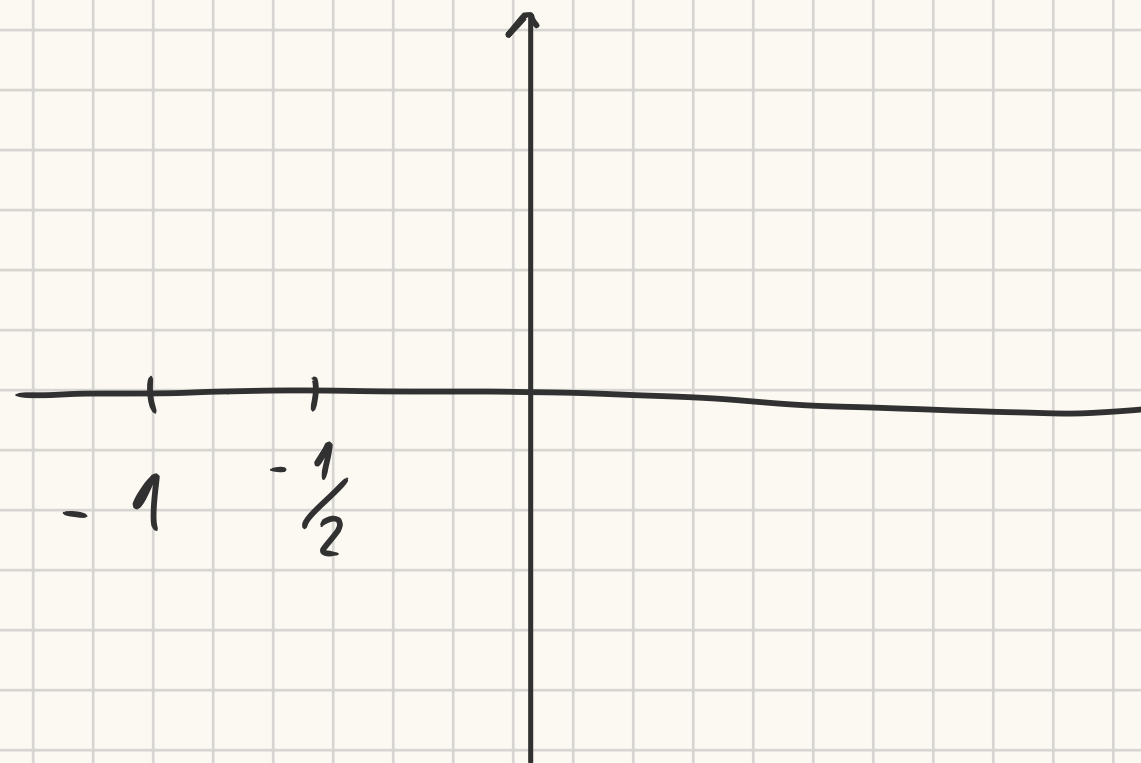
$$f(x) > g(x)$$

$$g(-\frac{2}{3}) = -\frac{2}{3} \cdot \frac{1}{\sqrt[3]{e}} = -\frac{2}{3\sqrt[3]{e}}$$

$$g(x) < 0 < f(x)$$

$$\int_{-1}^{-\frac{1}{2}} f(x) + g(x)$$

$$\int f(x) - g(x)$$



$$\int_{-1}^{-1/2} e^{2n+1} dn + \int_{-1}^{-1/2} n e^{2n+1} dn = e \left(\int_{-1}^{-1/2} e^{2n} dn + \int_{-1}^{-1/2} n e^{2n} dn \right)$$

$$= e \left(\left[\frac{1}{2} e^{2n} \right]_{-1}^{-1/2} + n \cdot \frac{e^{2n}}{2} \Big|_{-1}^{-1/2} - \int_{-1}^{-1/2} \frac{1}{2} e^{2n} dn \right) =$$

$$u = n \quad dv = e^{2n}$$

$$du = 1 \quad v = \frac{1}{2} e^{2n}$$

$$= \left[e \left(\frac{1}{2} (e^{-1} - e^{-2}) \right) \right] + \left[\frac{1}{2} \left(-\frac{1}{2} \cdot e^{-1} - (-1 \cdot e^{-2}) \right) - \frac{1}{4} (e^{-1} - e^{-2}) \right]$$

$$= e \left(\frac{1}{2} \left(\frac{1}{e} - \frac{1}{e^2} \right) + \frac{1}{2} \left(-\frac{1}{2e} + \frac{1}{e^2} \right) - \frac{1}{4} \left(\frac{1}{e} - \frac{1}{e^2} \right) \right)$$

$$= e \left(\frac{1}{2e} - \frac{1}{2e^2} - \frac{1}{4e} + \frac{1}{2e^2} - \frac{1}{4e} + \frac{1}{4e^2} \right) =$$

$$= \cancel{\frac{1}{2}} - \cancel{\frac{1}{2e}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{2e}} - \cancel{\frac{1}{4}} + \frac{1}{4e} = \frac{1}{4e}$$

$$\int_{-1}^{-1/2} x e^{2x+1} = \frac{1}{2} \int_{-1}^{-1/2} 2 e^{2x+1} = \left[\frac{1}{2} e^{2x+1} \right]_{-1}^{-1/2}$$

$$\frac{1}{2} (e^0 - e^{-1}) = \frac{1}{2} (1 - e^{-1}) = \frac{1}{2} \cdot 1 - \frac{1}{2} e^{-1} = \frac{1}{2} - \frac{e^{-1}}{2}$$

$$\int_{-1}^{-1/2} x e^{2x+1} = \left. \frac{e^{2x+1}}{2} \cdot x \right|_{-1}^{-1/2} - \int_{-1}^{-1/2} \frac{e^{2x+1}}{2} \cdot dx =$$

$u = x$
 $du = 1$

$dv = e^{2x+1}$
 $v = \frac{e^{2x+1}}{2}$

$$= \left. \frac{e^{2x+1}}{2} \cdot u \right|_{-1}^{-1/2} - \frac{1}{2} \int_{-1}^{-1/2} e^{2u+1} du = \frac{1}{2} \left(e^0 \cdot -\frac{1}{2} - \left(e^{-1} \cdot -1 \right) \right) - \frac{1}{2} \left[\frac{e^{2u+1}}{2} \right]_{-1}^{-1/2}$$

$$= \frac{1}{2} \left(-\frac{1}{2} + e^{-1} \right) - \frac{1}{4} \left(e^0 - e^{-1} \right) = \frac{1}{2} \left(-\frac{1}{2} + e^{-1} \right) - \frac{1}{4} \left(1 - e^{-1} \right)$$

$$= -\frac{1}{4} + \frac{e^{-1}}{2} - \frac{1}{4} + \frac{e^{-1}}{4} = -\frac{1}{2} + \frac{2e^{-1}}{4} + \frac{e^{-1}}{4} = -\frac{1}{2} + \frac{3e^{-1}}{4}$$

$$\frac{1}{2} - \frac{e^{-1}}{2} - \left(-\frac{1}{2} + \frac{3e^{-1}}{4} \right)$$

$$= \frac{1}{2} - \frac{e^{-1}}{2} + \frac{1}{2} - \frac{3e^{-1}}{4} = 1 - \frac{2e^{-1}}{4} - \frac{3e^{-1}}{4} = 1 - \frac{5e^{-1}}{4}$$

$$\int_a^b f(x) = - \int_b^a f(x)$$

Slide 25

① Se f é par $\int_{-a}^a f(u) du = 2 \int_0^a f(u) du$?

$$\int_{-a}^a f(u) du = \int_{-a}^0 f(u) du + \int_0^a f(u) du = - \int_0^a f(u) du + \int_0^a f(u) du \quad *$$

CA $\int_{-a}^a f(u) du = \int_0^a f(-u) \cdot (-dt)$

$$u = -t$$

$$du = -dt$$

u	t
$-a$	a
0	0

Como f é par

$$f(u) = f(-u)$$

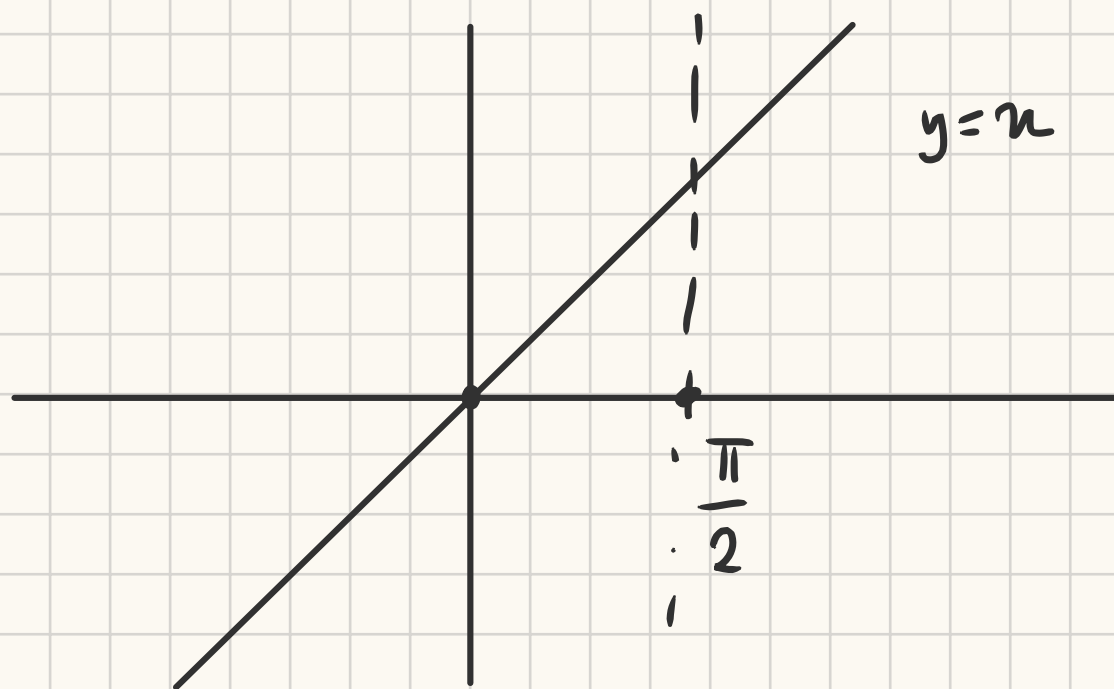
então:

$$\int_0^a f(-t) \cdot (-dt) = \int_0^a f(t) dt$$

$$\star \int_0^a f(u) \, du + \int_0^a f(t) \, dt = 2 \int_0^a f(u) \, du$$

Fiche 2.2

(23) Area entre $f(u) = u \cos(u)$, $y = u$, $u = 0$, $u = \frac{\pi}{2}$

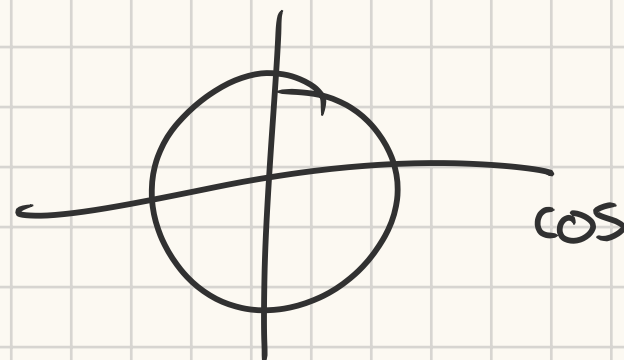


$$x = x \cos x \quad (\Rightarrow) \quad \cos x = 1 \quad (\Rightarrow) \quad x = 0$$

em $x = 0$)

$$f(0) = 0$$

$$y = 0$$



em $x = \frac{\pi}{2}$

$$f\left(\frac{\pi}{2}\right) = 0$$

$$y = \frac{\pi}{2}$$

$$\rightarrow y > f(x)$$

$$\int_0^{\pi/2} x - \int_0^{\pi/2} x \cos x = \left[\frac{x^2}{2} \right]_0^{\pi/2} - x \sin x \Big|_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx$$

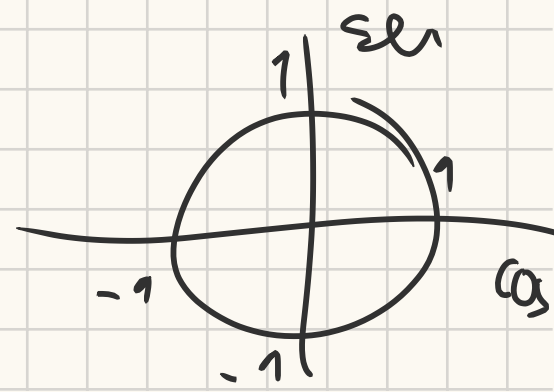
$$u = x$$

$$du = 1$$

$$dv = \cos x$$

$$v = \sin x$$

$$= \left[\frac{x^2}{2} \right]_0^{\pi/2} - \left[x \sin x \right]_0^{\pi/2} - \left[-\cos x \right]_0^{\pi/2} =$$



$$= \frac{\frac{\pi^2}{4}}{2} - 0 - \left(\frac{\pi}{2} - 0 \right) - (0 - 1) =$$

$$= \frac{\pi^2}{8} - \frac{\pi}{2} + 1 = \frac{\pi^2}{8} - \frac{4\pi}{8} + \frac{8}{8} = \frac{\pi^2 - 4\pi + 8}{8}$$

25!!

$$A = \{ (x, y) \in \mathbb{R}^2 : y \geq (x-3)^2, y \geq x-1, y \leq 4 \}$$

a) Representar a região

$$(x-3)^2 = x-1 \Leftrightarrow x^2 - 6x + 9 = x-1 \Leftrightarrow x^2 - 7x + 10 = 0 = (x=2) \vee (x=5)$$

$$x-1 = 4 \Leftrightarrow (x=5)$$

$$(x-3)^2 = 4 \Leftrightarrow x^2 - 6x + 9 = 4 \Leftrightarrow x^2 - 6x + 5 = 0 \Leftrightarrow (x=1) \vee (x=5)$$

C.A

$$n^2 - 7n + 10 = 0 \rightarrow n = \frac{7 \pm \sqrt{7^2 - 4 \cdot 10}}{2} \quad (\Rightarrow) \quad n = \frac{7 \pm \sqrt{49 - 40}}{2}$$

$$(\Rightarrow) n = \frac{7-3}{2} \vee n = \frac{7+3}{2} \quad (\Rightarrow) \quad n = \frac{4}{2} \vee n = \frac{10}{2} \quad (\Rightarrow) \quad n = 2 \vee n = 5$$

$$n^2 - 6n + 5 = 0 \rightarrow n = \frac{6 \pm \sqrt{36 - 20}}{2} \quad (\Rightarrow) \quad n = \frac{6-4}{2} \vee n = \frac{6+4}{2}$$

$$(\Rightarrow) n = \frac{2}{2} \vee n = \frac{10}{2} \quad (\Rightarrow) \quad n = 1 \vee n = 5$$

$$(n-3)^2 \quad \text{em} \quad n=0 \rightarrow (-3)^2 = 9$$

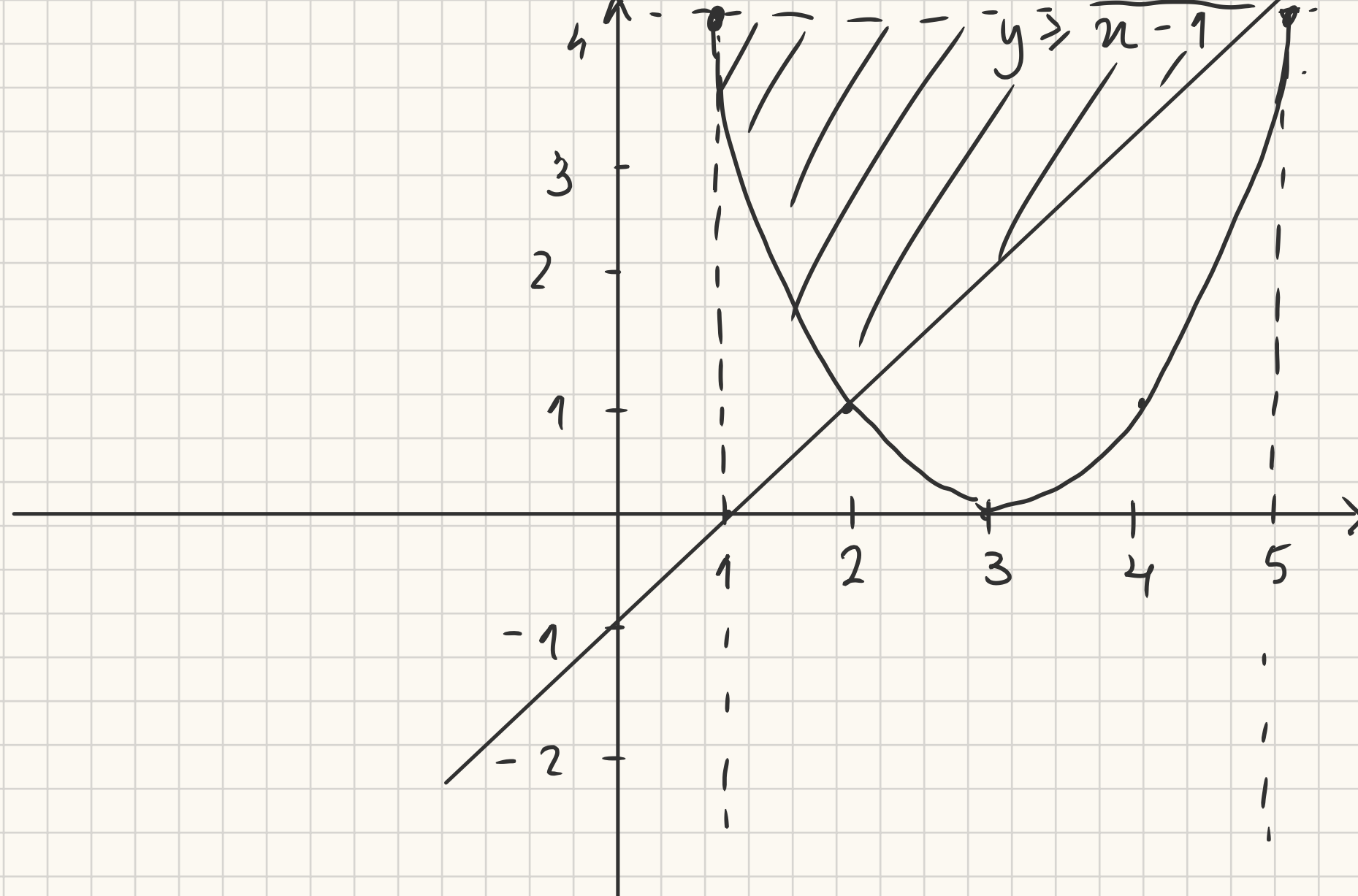
$$\text{em} \quad n=1 \rightarrow (-2)^2 = 4$$

$$\text{em} \quad n=2 \rightarrow (-1)^2 = 1$$

$$\text{em} \quad n=3 \rightarrow (0)^2 = 0$$

$$\text{em} \quad n=4 \rightarrow (1)^2 = 1$$

$$\text{em} \quad n=5 \rightarrow (2)^2 = 4$$



$$\int_1^2 4 - (x-3)^2 dx + \int_2^5 4 - x + 1 dx =$$

$$= \left[4x \right]_1^2 - \left[\frac{(x-3)^3}{3} \right]_1^2 + \left[4x \right]_2^5 - \left[\frac{x^2}{2} \right]_2^5 + \left[x \right]_2^5$$

$$= 8 - 4 - \left(\frac{(-1)^3}{3} - \frac{(-2)^3}{3} \right) + 20 - 8 - \left(\frac{25}{2} - \frac{4}{2} \right) + 5 - 2$$

$$= 4 + \frac{1}{3} - \frac{8}{3} + 12 - \frac{21}{2} + 3 = 19 - \frac{7}{3} - \frac{21}{2} =$$

$$= \frac{114}{6} - \frac{14}{6} - \frac{63}{6} = \frac{37}{6}$$

$$(29) \quad f(x) = \begin{cases} \frac{x}{(x^2+1)^{3/2}} \end{cases}$$

$$a) \int f(x) dx ?$$

$$\int \frac{u}{(u^2+1)^{3/2}} du = \int \frac{\sqrt{t-1}}{t^{3/2}} \cdot \frac{1}{2\sqrt{t-1}} dt = \int \frac{1}{2 t^{3/2}} dt$$

$$t = u^2 + 1 \Rightarrow \sqrt{t-1} = u$$

$$\frac{1}{2} (t-1)^{-1/2} = du$$

$$\frac{1}{2\sqrt{t-1}} = du$$

$$= \frac{1}{2} \int t^{-3/2} dt = \frac{1}{2} \frac{t^{-1/2}}{-1/2} = \frac{1}{2} \cdot \frac{1}{\sqrt{t}} \cdot (-2) = -\frac{1}{\sqrt{t}} + C = -\frac{1}{\sqrt{u^2+1}} + C, \quad C \in \mathbb{R}$$

b) Área de f , abscissas, $u = -1$ e $u = \sqrt{3}$?

$$f(u) = \frac{u}{(u^2+1)^{3/2}}$$

$$f(-1) = \frac{-1}{(1+1)^{3/2}} = \frac{-1}{2^{3/2}} = \frac{-1}{\sqrt{2^3}} = \frac{-1}{2\sqrt{2}}$$

$$f(u) = 0 \Leftrightarrow \frac{u}{(u^2+1)^{3/2}} = 0 \Leftrightarrow u = 0 \wedge (u^2+1)^{3/2} \neq 0 \quad \text{cond. Univ.}$$

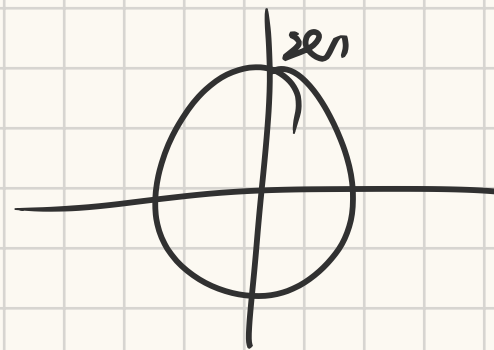
$$-\int_{-1}^0 f(u) + \int_0^{\sqrt{3}} f(u) = -\int_{-1}^0 \frac{u}{(u^2+1)^{3/2}} du + \int_0^{\sqrt{3}} \frac{u}{(u^2+1)^{3/2}} du$$

$$\text{c.A.} \int \frac{u}{(u^2+1)^{3/2}} du = \int \frac{u}{\sqrt{(u^2+1)^3}} du = \int \quad ? ?$$

Revisões Cap 1

Fiche 1

$$(35) \quad g(u) = \arcsen((u-1)^2)$$



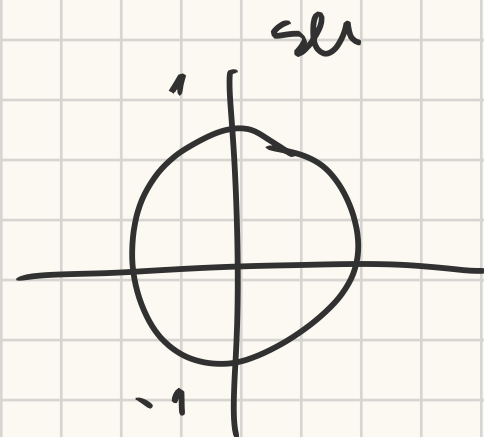
$$a) Dg = ?$$

$$Dg = \{ x \in \mathbb{R} : -1 \leq (x-1)^2 \leq 1 \}$$

$$-1 \leq (x-1)^2 \text{ cond univ } \wedge (x-1)^2 \geq 1 \Rightarrow x-1 \geq \pm 1$$

$$x-1 \leq 1 \wedge x-1 \geq -1 \Rightarrow x \leq 2 \wedge x \geq 0$$

$$Dg = [0, 2]$$



$$b) g(x) = \frac{\pi}{6} \text{ tem pelo menos 1 solu\c{c}\~ao em } [1, 2]$$

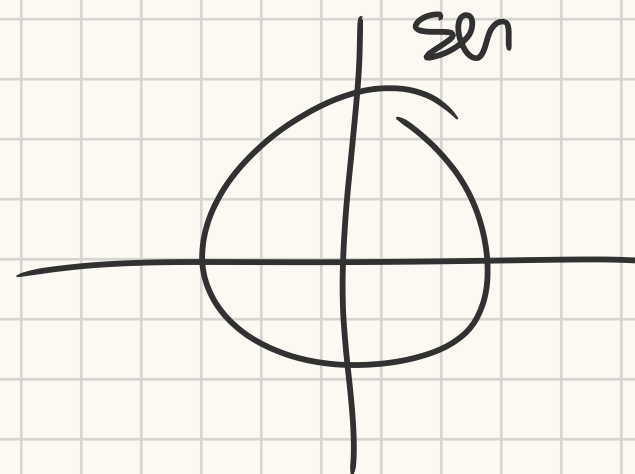
$$\arcsen((x-1)^2) = \frac{\pi}{6} \Leftrightarrow \arcsen((x-1)^2) - \frac{\pi}{6} = 0$$

$$\text{em } x=1)$$

$$\arcsen((0)^2) - \frac{\pi}{6} = -\frac{\pi}{6}$$

$$\text{em } x=2)$$

$$\arcsen((1)^2) - \frac{\pi}{6} = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{6}$$



$$\text{como } (g(1) - \frac{\pi}{6}) \cdot (g(2) - \frac{\pi}{6}) < 0 \text{ ent\~ao pelo T. Cauchy}$$

existe pelo menos uma solução em $[1, 2]$

c) Extremos locais e monotonia?

em $Dg = [0, 2]$

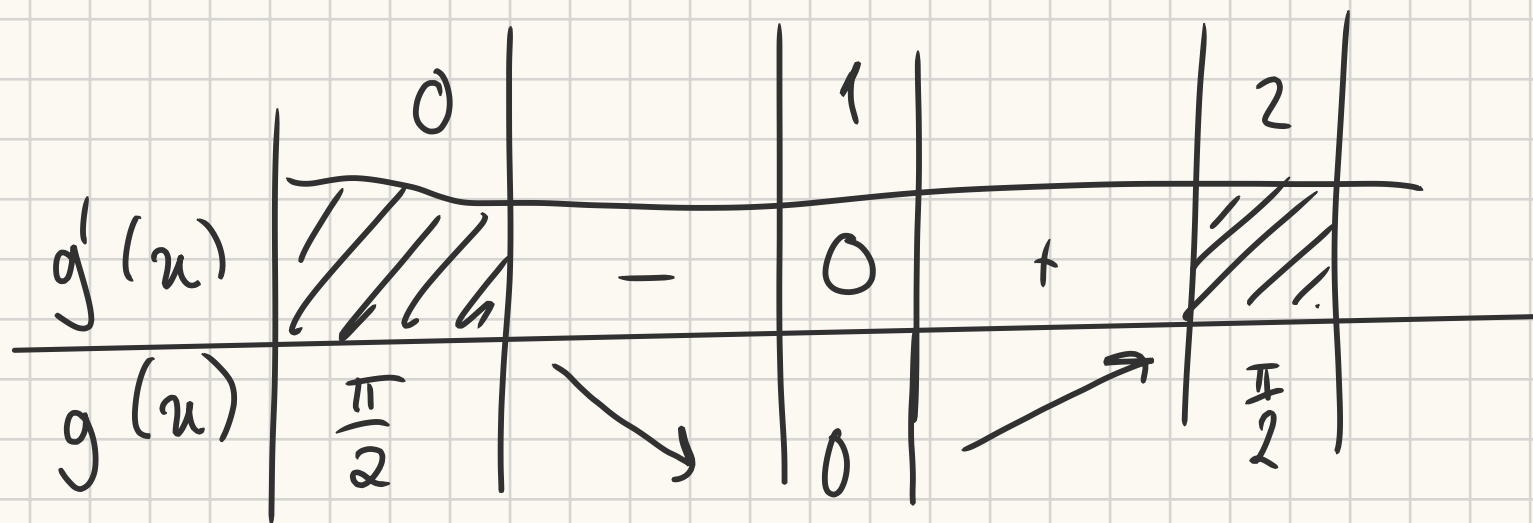
$$g'(u) = \arcsen((u-1)^2)' = \frac{((u-1)^2)'}{\sqrt{1-(u-1)^2}} = \frac{2u-2}{\sqrt{1-(u-1)^2}}$$

$$\frac{2u-2}{\sqrt{1-(u-1)^2}} = 0 \Leftrightarrow 2u-2=0 \wedge 1-(u-1)^2 > 0 \Leftrightarrow$$

$$\Leftrightarrow u=1 \wedge (u-1)^2 < 1 \Leftrightarrow$$

$$\Leftrightarrow u=1 \wedge u-1 < 1 \wedge u-1 > -1$$

$$\Leftrightarrow u=1 \wedge u < 2 \wedge u > 0$$



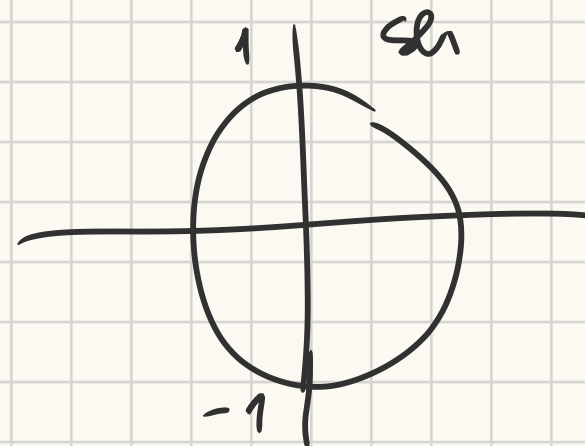
$$g'(\frac{1}{2}) = \frac{1-2}{\sqrt{1-(-\frac{1}{2})^2}} = \frac{-1}{\sqrt{1-\frac{1}{4}}}$$

$$g'(\frac{3}{2}) = \frac{2 \cdot \frac{3}{2} - 2}{\sqrt{1-(\frac{3}{2}-1)^2}} = \frac{1}{\sqrt{1-(\frac{1}{2})^2}}$$

$$g(0) = \arcsen((0-1)^2) = \arcsen(1) = \frac{\pi}{2}$$

$$g(1) = \arcsen((1-1)^2) = \arcsen(0) = 0$$

$$g(2) = \arcsen((2-1)^2) = \arcsen(1) = \frac{\pi}{2}$$



g é estritamente decrescente em $]0,1[$ e estritamente crescente em $]1,2[$
 g tem um mínimo em $x=1$, de valor 0

d) g é invertível?

$g(0) = g(2)$ logo g não é injetiva e por isso também não é invertível

Miniteste 1 2024/2025

① $f(x) = \operatorname{arccotg}(\ln(3x-1))$, D_f e $CD_f = ?$

$$D_f = \left\{ x \in \mathbb{R} : 3x - 1 > 0 \right\}$$

C.A

$$3x - 1 > 0 \Leftrightarrow 3x > 1 \Leftrightarrow x > \frac{1}{3}$$

$$D_f = \left] \frac{1}{3}, +\infty \right[$$

$$0 < \operatorname{arccotg}(x) < \pi$$

$$CD_f =]0, \pi[$$

$$0 < \operatorname{arccotg}(3x - 1) < \pi$$

$$0 < \operatorname{arccotg}(\ln(3x - 1)) < \pi$$



② $g(x) = \arccos(x - 2) + \sin(3 - x)$, $x \in [2, 3]$. Lagrange permette concludere che esiste un $c \in]2, 3[$ tal che $g'(c) = ?$

$$g'(c) = \frac{g(3) - g(2)}{3 - 2} = \frac{0 - \frac{\pi}{2} + \sin(1)}{1} = -\frac{\pi + 2\sin(1)}{2}$$

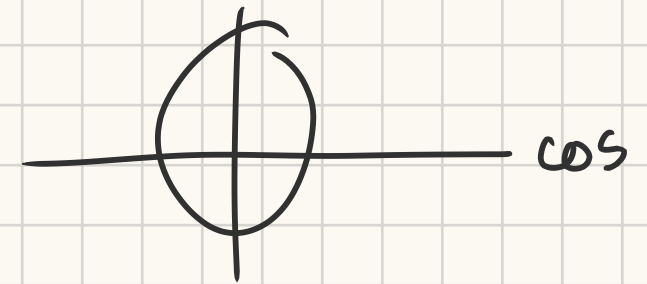
$$g(2) = \arccos(2 - 2) + \sin(3 - 2) = \arccos(0) + \sin(1) = \frac{\pi}{2} + \sin(1)$$

$$g(3) = \arccos(3 - 2) + \sin(3 - 3) = \arccos(1) + \sin(0) = 0$$

③ Seja h diferenciável tal que $h'(x) = 1 - e^{x^2 - 4}$. Número máximo de zeros de $h = ?$

$$h'(x) = 0 \Leftrightarrow 1 - e^{x^2 - 4} = 0 \Leftrightarrow e^{x^2 - 4} = 1 \Leftrightarrow x^2 - 4 = \ln(1) \Leftrightarrow x^2 - 4 = 0 \\ \Leftrightarrow x^2 = 4 \Leftrightarrow x = 2 \vee x = -2$$

3 zeros no máx



④ $\lim_{x \rightarrow 0} \frac{\lg x - \operatorname{sen} x}{x^2} = ?$

$$\lim_{x \rightarrow 0} \frac{\lg(x) - \operatorname{sen}(x)}{x^2} \stackrel{\text{Cauchy}}{=} \lim_{x \rightarrow 0} \frac{\sec^2 x - \cos x}{2x} \stackrel{\text{Cauchy}}{=} \lim_{x \rightarrow 0} \frac{(\cos^{\frac{1}{2}} x)' + \operatorname{sen} x}{2}$$

$$= \lim_{x \rightarrow 0} \frac{-2\cos(x)(-\operatorname{sen} x) + \operatorname{sen} x}{2} = \frac{0}{2} = 0$$

$$\textcircled{5} \quad T_0^2 f(x) = ? \quad , \quad f(x) = \cos x \quad , \quad \cos\left(\frac{1}{2}\right) = ?$$

$$\begin{array}{ll} f(x) = \cos x & f(0) = 1 \\ f'(x) = -\sin x & f'(0) = 0 \\ f''(x) = -\cos x & f''(0) = -1 \end{array}$$

$$\begin{aligned} T_0^2 f(x) &= \frac{1}{0!} (x-0)^0 + \frac{0}{1!} (x-0)^1 + \frac{-1}{2!} (x-0)^2 \\ &= 1 + 0 - \frac{1}{2} x^2 = 1 - \frac{x^2}{2} \end{aligned}$$

$$\cos x = \cos\left(\frac{1}{2}\right) \Leftrightarrow x = \frac{1}{2} \rightarrow 1 - \frac{\left(\frac{1}{2}\right)^2}{2} = 1 - \frac{\left(\frac{1}{4}\right)}{2} = 1 - \frac{1}{8} = \frac{7}{8}$$

Teste 1 - 202

$\textcircled{1}$ a)

$$f(a) < 0 \quad \text{e} \quad f(b) < 0 \quad , \quad a < b$$

se $f(c) > 0$ então existem pelo menos dois zeros em $]a, b$

$$b) \varphi(x) = g(1-x^2), \quad g: [0, +\infty[\rightarrow \mathbb{R} \text{ continua}$$

$$D\varphi = \{x \in \mathbb{R} : 1-x^2 \geq 0\} \quad D\varphi = [-1, 1]$$

CA.

$$1-x^2 \geq 0 \Leftrightarrow x^2 \leq 1 \Leftrightarrow x \geq -1 \wedge x \leq 1$$

Como $D\varphi$ é fechado e limitado, pelo T. Weierstrass φ tem máximos e mínimos globais nesse domínio

$$c) h(x) = (x+1) \arcsen(\sqrt{x}-1), \quad x \in [0, 4]. \quad c \in]0, 4[. \quad h'(c) = ?$$

$$h'(c) = \frac{h(4) - h(0)}{4 - 0} = \frac{\frac{5\pi}{2} + \frac{\pi}{2}}{4} = \frac{6\pi}{2} \cdot \frac{1}{4} = \frac{3\pi}{4}$$

$$h(4) = 5 \arcsen(2-1) = 5 \arcsen(1) = \frac{5\pi}{2}$$

$$h(0) = \arcsen(-1) = -\frac{\pi}{2}$$

d) $g(n) = n^{21} - n + n$. Max zeros de g ?

$g'(n) = 21n^{20} - 1$: g' tem 2 zeros $\rightarrow g$ tem no max 3

e) $\lim_{n \rightarrow 0} (n^4 + 1)^{1/n^2} = ?$

$$\begin{aligned} \lim_{n \rightarrow 0} (n^4 + 1)^{1/n^2} &= \lim_{n \rightarrow 0} e^{\ln(n^4 + 1)^{1/n^2}} = \lim_{n \rightarrow 0} e^{\frac{1}{n^2} \ln(n^4 + 1)} = \\ &= e^{\lim_{n \rightarrow 0} \frac{\ln(n^4 + 1)}{n^2}} = e^{\lim_{n \rightarrow 0} \left(\frac{\frac{4n^3}{n^4 + 1}}{1} \right)} = e^{\lim_{n \rightarrow 0} \left(\frac{4n^3}{n^4 + 1} \right)} = e^{\frac{0}{1}} = e^0 = 1 \end{aligned}$$

f) $n \in [2, 4]$ $G(n) = \int_{2n+1}^2 \ln t \, dt$. $G'(n) = ?$

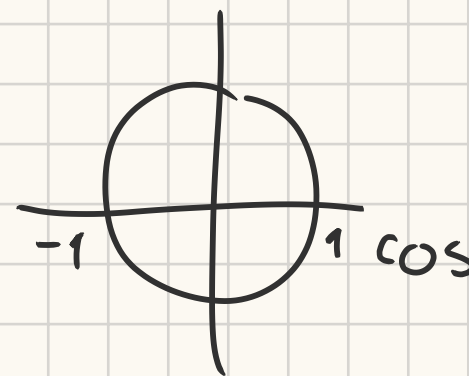
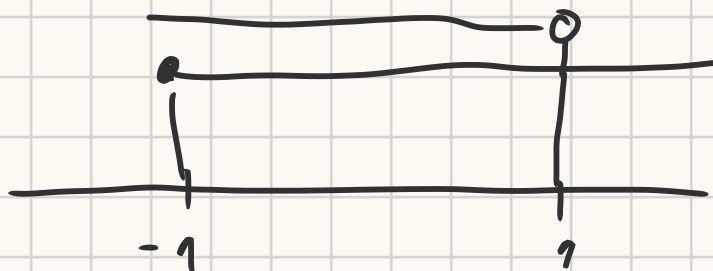
$G'(n) = \ln(2) \cdot (2)' - \ln(2n+1) \cdot (2n+1)' = 0 - \ln(2n+1) \cdot 2 = -2\ln(2n+1)$

② $f(x) = \ln(\arccos(x))$, $f^{-1}=?$, $D_{f^{-1}}=?$, $CD_{f^{-1}}=?$

$$D_f = \left\{ x \in \mathbb{R} : -1 \leq x \leq 1 \wedge \arccos(x) > 0 \right\}$$

C.A

- $x \geq -1$
- $x \leq 1$



• $\arccos(x) > 0 \Leftrightarrow x < \cos(0) \Leftrightarrow x < 1$ $D_f = [-1, 1[= CD_{f^{-1}}$

CD_f

$$]-\infty, 0[\cup \ln(x) <]0, +\infty[$$

$$0 \leq \arccos(x) \leq \pi$$

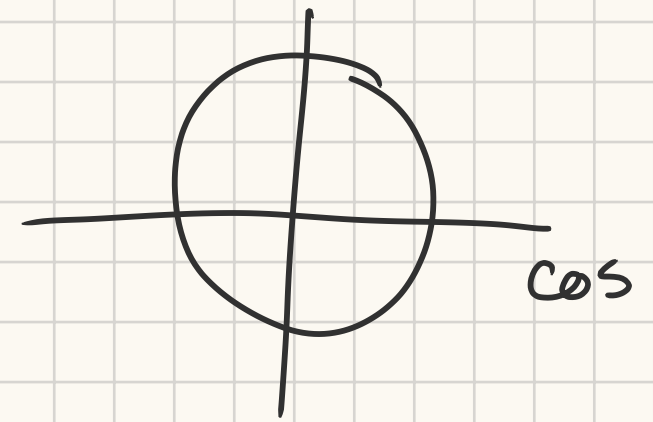
$$0 < \ln(\arccos(x)) \leq \ln(\pi)$$

$$\ln(0) \text{ até } \ln(\pi)$$

$$CD_f =]0, \ln(\pi)] = D_{f^{-1}}$$

f^{-1}) $y = \ln(\arccos(x)) \Leftrightarrow e^y = \arccos(x) \Leftrightarrow \cos(e^y) = x$

$$f^{-1}(u) = \cos(e^u)$$



b) reta tangente a f^{-1} em $u = \ln\left(\frac{\pi}{2}\right)$

$$(f^{-1})' = (\cos(e^u))' = -e^u \operatorname{sen}(e^u)$$

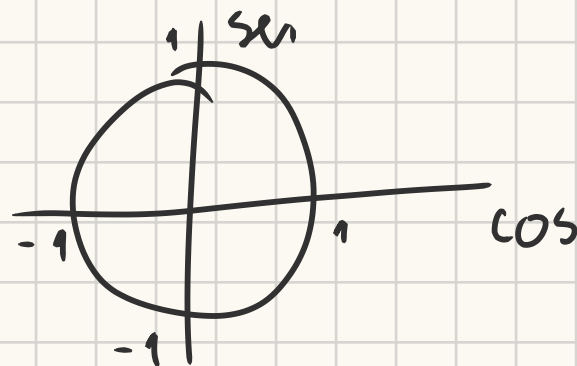
$$(f^{-1})'\left(\ln\frac{\pi}{2}\right) = -e^{\ln\frac{\pi}{2}} \operatorname{sen}(e^{\ln\frac{\pi}{2}}) = -\frac{\pi}{2} \cdot \operatorname{sen}\left(\frac{\pi}{2}\right) = 1 \cdot -\frac{\pi}{2} = -\frac{\pi}{2} = m$$

$$f^{-1}\left(\ln\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$

$$y = mn + b \Leftrightarrow 0 = -\frac{\pi}{2} \ln\left(\frac{\pi}{2}\right) + b$$

$$\Leftrightarrow \frac{\pi}{2} \ln\left(\frac{\pi}{2}\right) = b$$

$$y = \frac{\pi}{2} \left(-u + \ln\left(\frac{\pi}{2}\right)\right) \Leftrightarrow y = \frac{\pi}{2} \left(\ln\left(\frac{\pi}{2}\right) - u\right)$$



$$(3) \quad g(x) = x - \cos(x) \quad . \quad T_{\pi}^3 g = ?$$

$$g(x) = x - \cos(x)$$

$$g'(x) = 1 + \sin(x)$$

$$g''(x) = \cos(x)$$

$$g'''(x) = -\sin(x)$$

$$g(\pi) = \pi - \cos(\pi) = \pi + 1$$

$$g'(\pi) = 1 + \sin(\pi) = 1$$

$$g''(\pi) = \cos(\pi) = -1$$

$$g'''(\pi) = 0$$

$$T_{\pi}^3 g = \frac{\pi+1}{0!} (x-\pi)^0 + \frac{1}{1!} (x-\pi)^1 + \frac{-1}{2!} (x-\pi)^2 + \frac{0}{3!} (x-\pi)^3$$

$$= \pi + 1 + x - \pi - \frac{1}{2} (x-\pi)^2 = 1 + x - \frac{1}{2} (x-\pi)^2$$

$$g^{IV}(x) = -\cos(x)$$

$$R_{\pi}^3 = \frac{-\cos(\theta)}{4!} (x-\pi)^4$$

$$x = 3, \quad c = \pi \rightarrow \theta \in]3, \pi[$$

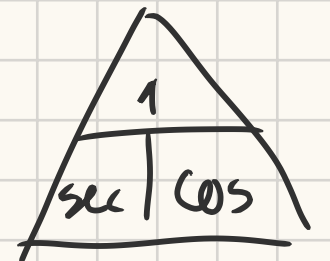
$$\cos(\pi) = -1$$

$$\begin{aligned}
 |R_\pi^3| &= \left| \frac{\cos(\theta)}{8} \right| \underbrace{|(3-\pi)^4|}_{>0} = \left| \frac{\cos(\theta)}{8} \right| (3-\pi)^4 < \left| \frac{\cos(\pi)}{8} \right| (3-\pi)^4 \\
 &< \left| -\frac{1}{8} \right| (3-\pi)^4 \\
 &< \frac{(3-\pi)^4}{8}
 \end{aligned}$$

$$(4) \int \frac{1}{n^2 \sqrt{n^2 - 3}} \, dn \quad (\text{use } n = \sqrt{3} \sec t)$$

$$n = \sqrt{3} \sec(t) \quad \Rightarrow \quad dn = \sqrt{3} \sec(t) \cdot \tan(t) \quad \Rightarrow \quad \frac{n}{\sqrt{3}} = \sec t \quad \Rightarrow$$

$$\int \frac{1}{3 \sec^2 t \sqrt{3 \sec^2 t - 3}} \cdot \sqrt{3} \sec(t) \cdot \tan(t) \, dt =$$



$$= \int \frac{\sqrt{3} \sec(t) \cdot \tan(t)}{3 \sec^2 t \sqrt{3 \tan^2 t}} \, dt = \int \frac{\cancel{\sqrt{3}} \cancel{\sec(t)} \cdot \cancel{\tan(t)}}{3 \sec^2(t) \cdot \cancel{\tan(t)} \cdot \cancel{\sqrt{3}}} \, dt = \frac{1}{3} \int \cos(t) \, dt$$

$$= \frac{1}{3} \sin(t) + C = *$$

$$\sec(t) = \frac{x}{\sqrt{3}}$$

$$\sin(t) = ?$$

$$\frac{x}{\sqrt{3}} = \frac{1}{\cos t} \Rightarrow \cos t = \frac{\sqrt{3}}{x}$$

$$\cos^2 t + \sin^2 t = 1$$

$$\Rightarrow \sin^2 t = 1 - \cos^2 t$$

$$\Rightarrow \sin^2 t = 1 - \frac{3}{x^2}$$

$$\Rightarrow \sin t = \sqrt{\frac{x^2 - 3}{x^2}} \quad \Rightarrow \sin t = \frac{\sqrt{x^2 - 3}}{x}$$

$$* \frac{1}{3} \sin t + C = \frac{1}{3} \frac{\sqrt{x^2 - 3}}{x} + C, \quad C \in \mathbb{R}$$

$$b) \int \frac{x-2}{(1+x^2)(x+3)} dx$$

$$\frac{n-2}{(1+n^2)(n+3)} = \frac{An+B}{(1+n^2)} + \frac{C}{(n+3)} = \frac{A(n^2+3n) + B(n+3) + C(n^2+1)}{(1+n^2)(n+3)}$$

$$= \frac{A+C(n^2) + (B+3A)(n) + (3B+C)(1)}{(1+n^2)(n+3)}$$

$$\begin{cases} A+C=0 \\ B+3A=1 \\ 3B+C=-2 \end{cases} \quad \begin{cases} A=-C \\ C=-2-3B \end{cases} \quad \begin{cases} A=2+3B \quad (\Rightarrow) \quad A-2=3B \\ \frac{A-2}{3} + 3A=1 \end{cases} \quad \begin{cases} \frac{1}{2}-2=3B \\ A=\frac{1}{2} \end{cases}$$

$$\frac{A-2}{3} = 1-3A \quad (\Rightarrow) \quad A-2 = 3-9A \quad (\Rightarrow)$$

$$(\Rightarrow) A = 3-9A+2 \quad (\Leftrightarrow) \quad A+9A = 5$$

$$(\Rightarrow) 10A = 5 \quad (\Rightarrow) \quad A = \frac{1}{2}$$

$$\frac{1}{2} - \frac{4}{2} = 3B \quad (\Rightarrow) \quad -\frac{3}{2} = 3B$$

$$(\Rightarrow) -\frac{3}{2} \cdot \frac{1}{3} = B \quad (\Rightarrow) \quad B = -\frac{1}{2}$$

$$\begin{cases} B = -\frac{1}{2} \\ A = \frac{1}{2} \\ C = -2 + \frac{3}{2} \end{cases} \quad \begin{cases} B = -\frac{1}{2} \\ A = \frac{1}{2} \\ C = -\frac{1}{2} \end{cases}$$

$$\int \frac{x-2}{(1+x^2)(x+3)} = \int \frac{\frac{1}{2}(x) + (-\frac{1}{2})}{(1+x^2)} + \frac{-\frac{1}{2}}{(x+3)} dx$$

$$= \int \frac{x-1}{2} \cdot \frac{1}{(1+x^2)} + \left(-\frac{1}{2}\right) \cdot \frac{1}{x+3} dx = \frac{1}{2} \int \frac{x-1}{x^2+1} dx - \frac{1}{2} \int \frac{1}{x+3} dx$$

$$= \frac{1}{2} \left(\int \frac{x}{x^2+1} dx - \int \frac{1}{x^2+1} dx - \int \frac{1}{x+3} dx \right) = *$$

C.A $\int \frac{x}{x^2+1} dx \longrightarrow \int \frac{\frac{\sqrt{t}-1}{t}}{2\sqrt{t-1}} dt$

$$t = x^2 + 1 \quad (\Rightarrow) \quad \sqrt{t-1} = x$$

$$\frac{1}{2} (t-1)^{-\frac{1}{2}} = dx$$

$$\Rightarrow \frac{1}{2\sqrt{t-1}} dt = dx$$

$$= \int \frac{1}{2t} dt = \frac{1}{2} \int \frac{1}{t} dt = \ln |t| = \frac{1}{2} \ln |x^2 + 1|$$

$$* \frac{1}{2} \left(\frac{1}{2} \ln |x^2 + 1| - \operatorname{arctg}(x) - \ln |x + 3| \right) = - \frac{\ln |x + 3|}{2} + \frac{\ln |x^2 + 1|}{4} - \frac{\operatorname{arctg}(x)}{2}$$

⑤ $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} |\operatorname{arctg}(x)| & , x \leq 0 \\ \operatorname{sen}(2x) & , x > 0 \end{cases}$$

a) $\int \operatorname{arctg}(x) dx = ?$

$$\int \operatorname{arctg}(x) dx = \int 1 \cdot \operatorname{arctg}(x) dx = \text{Substituição}$$

$$u = \operatorname{arctg}(x) \quad du = \frac{1}{1+x^2} dx$$

$$dv = 1 \quad v = x$$

$$= \operatorname{arctg}(x) \cdot x - \int x \frac{1}{1+x^2} dx = \operatorname{arctg}(x) \cdot x - \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= \arctg(u) \cdot u - \frac{1}{2} \ln|1+u^2| + C, \quad C \in \mathbb{R}$$

b) f é integrável em $[-1, 1]$? $\int_{-1}^1 f(u) = ?$

$$\lim_{u \rightarrow 0^-} f(u) = \lim_{u \rightarrow 0^-} |\arctg(u)| = 0$$

$$\lim_{u \rightarrow 0^+} f(u) = \lim_{u \rightarrow 0^+} \sin(2u) = 0$$

f é contínua em $[-1, 1]$, logo é integrável em $[-1, 1]$

$$f(0) = |\arctg(0)| = 0$$

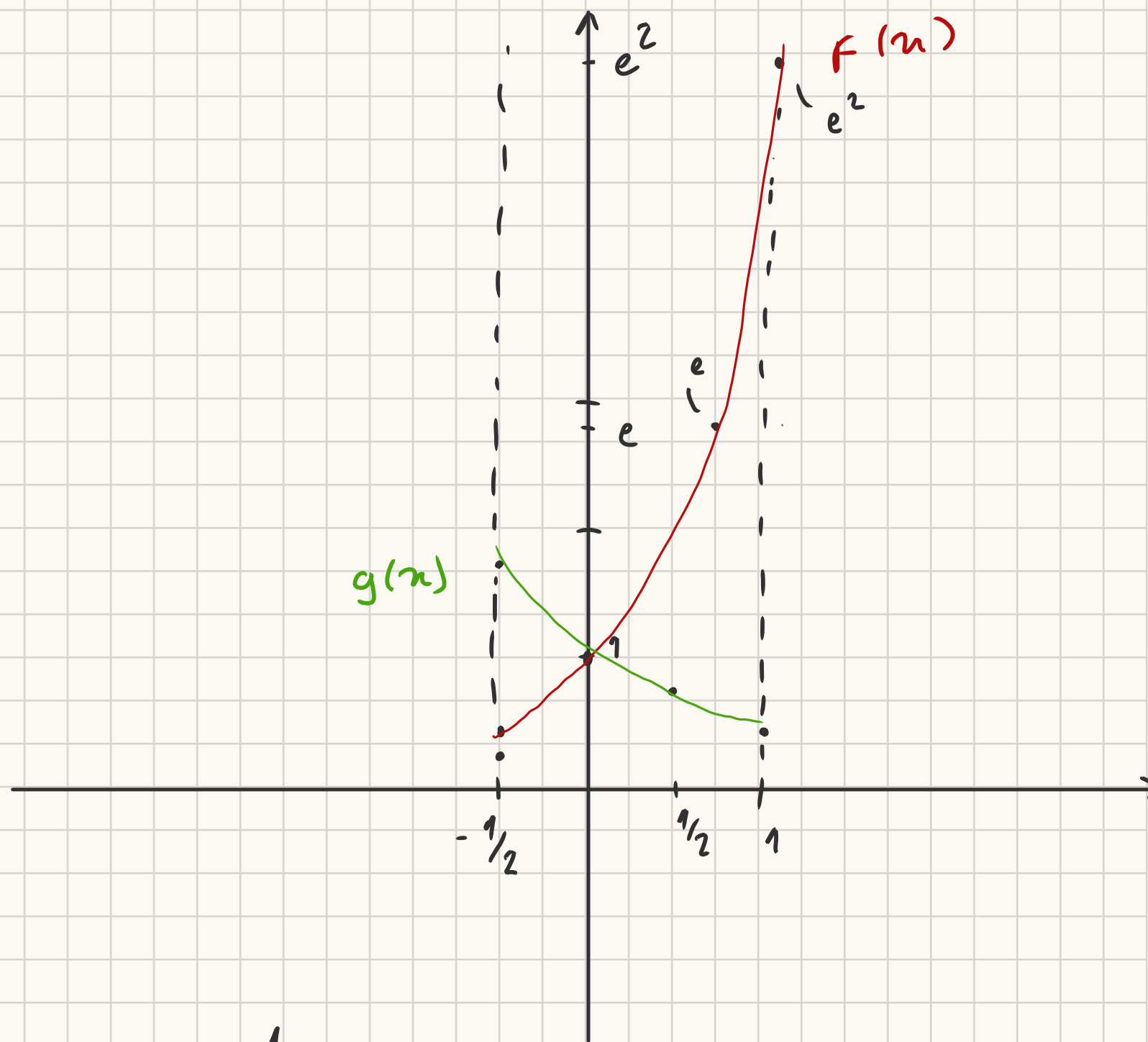
$$\int_{-1}^1 f(u) = \int_{-1}^0 |\arctg(u)| \, du + \int_0^1 \sin(2u) \, du$$

$$\begin{aligned}
&= \left[\arctg(u) \cdot u - \frac{\ln|1+u^2|}{2} \right]_{-1}^0 + \frac{1}{2} \left[-\cos(2u) \right]_0^1 \\
&= \left[\arctg(|u|) \cdot |u| - \frac{\ln(1+u^2)}{2} \right]_{-1}^0 + \frac{1}{2} \left[-\cos(2u) \right]_0^1 \\
&= \arctg(0) \cdot 0 - \frac{\ln(1)}{2} - \left(\arctg(|-1|) \cdot |-1| - \frac{\ln(1+(-1)^2)}{2} \right) + \frac{1}{2} (-\cos(2) - (-\cos(0))) \\
&= 0 - \left(\arctg(1) \cdot 1 - \frac{\ln(2)}{2} \right) + \frac{1}{2} (-\cos(2) + 1) = -\left(\frac{\pi}{4} - \frac{\ln(2)}{2} \right) - \frac{\cos(2) - 1}{2} \\
&= \frac{\pi}{4} + \frac{\ln(2)}{2} - \frac{\cos(2) - 1}{2} = \frac{\pi}{4} + \frac{2\ln(2)}{4} - \frac{2\cos(2) - 2}{4} = \frac{\pi + \ln(4) - 2\cos(2) + 2}{4} + C \\
&\quad , C \in \mathbb{R}
\end{aligned}$$

⑥ $f(u) = e^{2u}$, $g(u) = e^{-u}$. Esboce região delimitada por $f, g, u = -\frac{1}{2}, u = 1, 0$

$$\begin{aligned}
f(1) &= e^2 & g(1) &= e^{-1} = \frac{1}{e} > \frac{1}{3} \\
f(1/2) &= e^{2 \cdot 1/2} = e^1 = e & g(1/2) &= e^{-1/2} = \frac{1}{\sqrt{e}} > \frac{1}{\sqrt{3}} > \frac{1}{3} \\
f(0) &= e^0 = 1 & g(0) &= e^{-0} = 1 \\
f(-1/2) &= e^{-1/2 \cdot 2} = e^{-1} = \frac{1}{e} & g(-1/2) &= e^{1/2} = \sqrt{e}
\end{aligned}$$

$$\begin{aligned}
e^{2u} &= e^{-u} \quad (\Rightarrow) \quad 2u = -u \\
(\Rightarrow) \quad 2u + u &= 0 \quad (\Rightarrow) \quad 3u = 0 \\
(\Rightarrow) \quad u &= 0
\end{aligned}$$



$$\int_{-1/2}^0 g(x) - f(x) + \int_0^1 f(x) - g(x)$$

$$= \int_{-1/2}^0 e^{-u} - e^{2u} du + \int_0^1 e^{2u} - e^{-u} du =$$

$$- \int_{-1/2}^0 -e^{-u} du - \frac{1}{2} \int_{-1/2}^0 2e^{2u} du + \frac{1}{2} \int_0^1 2e^{2u} du - \int_0^1 -e^{-u} du$$

$$= - \left[e^{-u} \right]_{-1/2}^0 - \frac{1}{2} \left[e^{2u} \right]_{-1/2}^0 + \frac{1}{2} \left[e^{2u} \right]_0^1 - \left[e^{-u} \right]_0^1$$

$$- (e^{-0} - e^{+1/2}) - \frac{1}{2} (e^{2 \cdot 0} - e^{2 \cdot -1/2}) + \frac{1}{2} (e^{2 \cdot 1} - e^{2 \cdot 0}) - (e^{-1} - e^{-0})$$

$$= -1 + \sqrt{e} - \frac{1}{2} + \cancel{\frac{1}{e}} + \frac{1}{2} e^{-1} + \frac{1}{2} e^2 - 1 - e^{-1} + \cancel{1}$$

$$= e^{1/2} + \frac{e^{-1}}{2} + \frac{e^2}{2} - e^{-1} - \frac{1}{2} - 1 = \frac{2\sqrt{e} + e^{-1} + e^2 - 2e^{-1} - 3}{2} = \frac{2\sqrt{e} + e^2 - 2e^{-1} - 3}{2e}$$

$$= 2$$

$$\lim (\dots) = \lim e^{\ln(\dots)} = e^{\lim \ln(\dots)}$$

$$a) \int \frac{1}{x^3 + 8} dx = \int \frac{1}{(x - (-2))(x^2 - 2x + 4)}$$

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & 8 \\ -2 & & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & 0 \end{array}$$

$$\frac{1}{(x+2)(x^2-2x+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2-2x+4} = \frac{A(x^2-2x+4) + B(x^2+2x) + C(x+2)}{(x+2)(x^2-2x+4)}$$

$$\begin{cases} A + B = 0 \\ -2A + 2B + C = 0 \\ 4A + 2C = 1 \end{cases}$$

$$\begin{cases} A = -B \\ -2A - 2A = -C \end{cases}$$

$$\begin{cases} -4A = -C \\ 3C = 1 \end{cases}$$

$$\begin{cases} -4A = -1/3 \\ C = 1/3 \end{cases}$$

$$\begin{cases} B = -1/12 \\ A = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12} \\ C = 1/3 \end{cases}$$

$$\int \frac{1}{(x+2)(x^2-2x+4)} dx = \int \frac{1/12}{x+2} dx + \int \frac{(-1/12)x + 1/3}{(x^2-2x+4)} dx$$

$$= \frac{1}{12} \int \frac{1}{x+2} dx - \frac{1}{12} \int \frac{x-4}{x^2-2x+4} dx$$

$$= \frac{1}{12} \ln|x+2| - \frac{1}{12} \left(\int \frac{x}{x^2-2x+4} dx - \int \frac{4}{x^2-2x+4} dx \right)$$

$$= \quad // \quad -$$